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Advanced Analytics in Business

ASSIGNMENTS GROUP 10

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Chapter 1

Assignment 1: Predictive Modeling of Tabular Data

1.1 Intro

This project aims to develop a predictive model for the revenue of a European fashion webshop, specifically focusing on online shoe sales. After data exploration and cleaning, features were engineered for use in various models. Performance is evaluated using Mean Absolute Error (MAE) and Spearman Correlation.

1.2 Data

The dataset comprises three core files reflecting the modeling pipeline:

- **Transactions (2016–2017):** Detailed customer transaction data used as the basis for feature engineering.
- **Training Labels (2018–2019):** Revenue data for the training set, linking customer IDs to actual revenue during the target period.
- **Test Set:** Unique customer identifiers for whom future revenue must be predicted.

1.2.1 Train, Validation and Test split

The data was partitioned using an out-of-sample split. Customers were randomly assigned to two groups to ensure the model learns generalizable patterns rather than specific indi-

vidual behavior:

1. **Training Set:** Buying behavior (2016–2017) and corresponding revenue (2018–2019).
2. **Test Set:** Buying behavior (2016–2017) only.

A 20% validation set was created using a stratified split. Because 63% of customers generated no revenue (zero-inflation), we stratified by a binary indicator (revenue > 0 vs. revenue = 0). This ensures both training and validation sets maintain identical proportions of active versus inactive customers.

1.2.2 Data Exploration

Before any data preparation and modeling, the data was explored to gain a deeper understanding of the data at hand. The dataset contains 27 attributes for 344,212 transactions. These include order/pack dates, sale IDs, discounts, revenue, and product characteristics such as brand, color, and material. Table 1.1 provides an overview.

Table 1.1: Overview of Attributes and Characteristics

Variable	Measurement Level	% Miss.	Unique Values / Statistics	Description
cust_id	Nominal	0.00%	145,739	Unique Customer ID
sale_id	Nominal	0.00%	245,928	Transaction ID
returned_to_shop_id	Nominal	81.43%	78	Return location
prod_title	Nominal	0.00%	21,302	Product name
prod_id	Nominal	0.00%	22,324	Product ID
prod_season	Nominal	0.00%	23	Season code
prod_brand	Nominal	0.00%	493	Brand name
prod_color	Nominal	0.00%	41	Primary color
prod_type_1	Nominal	0.00%	4	Target group (e.g., Women)
prod_type_3	Nominal	0.00%	41	Category (e.g., Sneaker)
prod_type_4	Nominal	58.61%	48	Sub-style classification
prod_type_5	Nominal	1.82%	51	Additional style descriptor
prod_heel	Nominal	38.41%	4	Heel type
prod_material	Nominal	6.61%	58	Main outer material
prod_print	Nominal	77.49%	128	Pattern or print type
prod_comfort_sole	Nominal	87.76%	51	Comfort sole indicator
prod_comfort_wear	Nominal	90.55%	13	Enhanced comfort fit
prod_clasp	Nominal	11.23%	48	Closing mechanism (Lace, Zip)
prod_web_only	Binary	0.00%	2	Online exclusive indicator
prod_insole	Binary	7.54%	2	Insole feature indicator
prod_outlet	Binary	0.00%	2	Sold via outlet channel
prod_size	Ordinal	0.00%	min: 16, max: 50, mode: 39	Shoe size
sale_discount	Numeric	0.00%	mean: 20, std: 22.48	Discount value in EUR
sale_revenue	Numeric	0.00%	mean: 53.07, std: 38.11	Revenue per line item
order_date	Date	0.00%	01/01/16 – 12/31/17	Order placement date
pack_date	Date	0.00%	01/03/16 – 01/12/18	Shipping/Packing date

Target Variable

A very high amount of people (63,41%) have a revenue equal to 0. Due to the heavy right-skew and zero-inflation of the revenue data (see Figure 1.1), a logarithmic transformation ($\log_{1p}$) was applied. This visualization reveals a bimodal distribution, clearly separating the non-returning customers (peak at 0) from the active customer base.

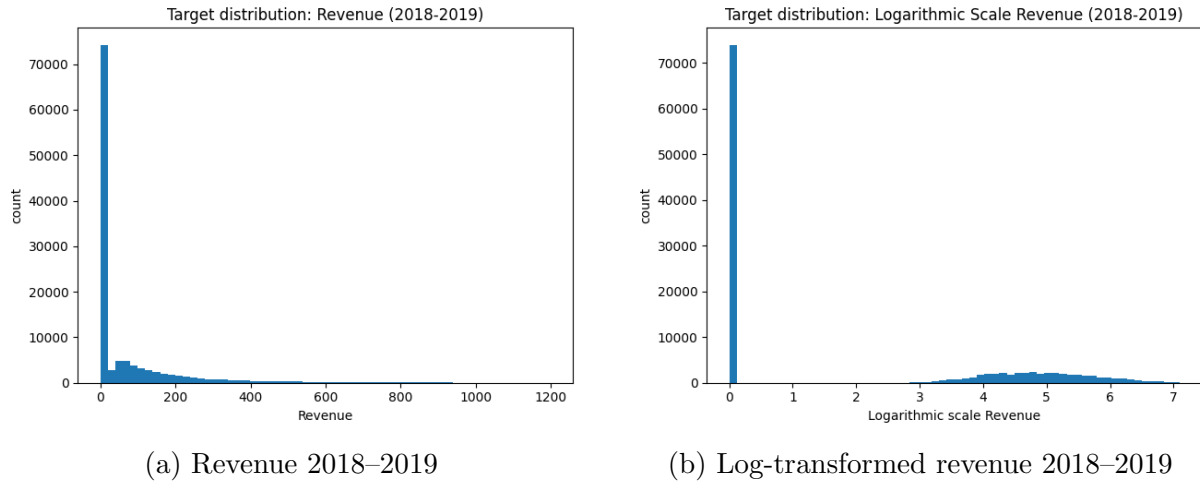


Figure 1.1: Distribution of the training set revenue before and after log transformation.

Correlation Matrix

To better understand the relationships between the numerical features, a correlation matrix was computed. This analysis was used as an exploratory step to identify groups of variables that carry similar information and to assess the presence of multicollinearity in the dataset.

As shown in Figure 1.2, several blocks of correlated variables can be observed. For example, transaction-volume features such as `n_items`, `n_returns`, `n_sales_non_returned`, and `total_discount` are strongly related. Similar patterns can also be observed for size-related variables, discount-related variables, and purchase-frequency indicators. This suggests that part of the feature space contains overlapping information.

Although this multicollinearity was not directly corrected at this stage, since the final machine learning models are mostly tree-based and can generally handle correlated predictors, the correlation matrix motivated a further exploratory analysis using Principal Component Analysis (PCA). The goal was therefore not to use PCA as a preprocessing step, but to better understand the dimensional structure of the data.

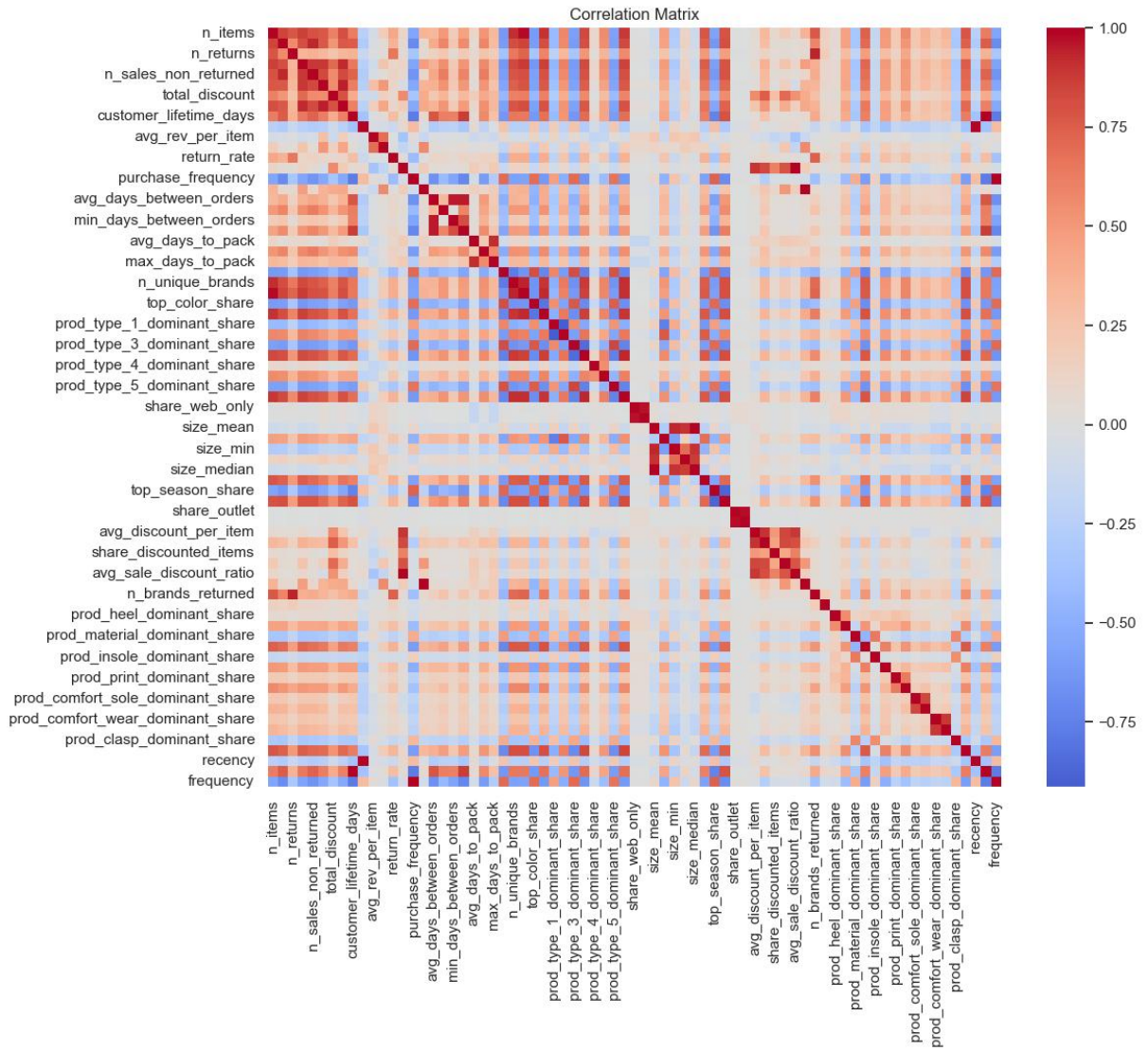


Figure 1.2: Correlation matrix of the numerical features.

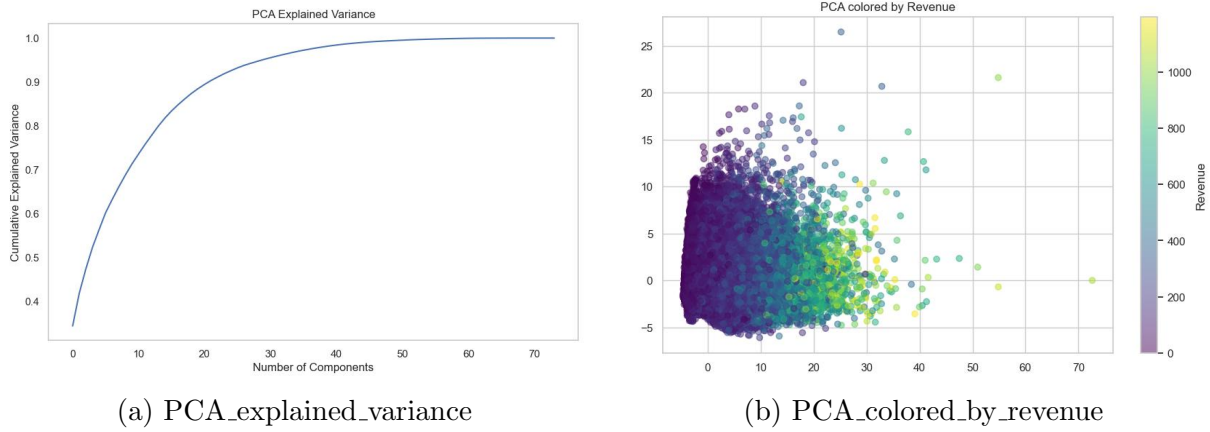


Figure 1.3: PCA results.

Principal Component Analysis

Following the correlation analysis, a Principal Component Analysis (PCA) was conducted as an additional exploratory step. Since the correlation matrix indicated the presence of overlapping information between several features, PCA was used to investigate whether the variation in the data could be summarized using a smaller number of dimensions.

Figure 1.3(a) presents the cumulative explained variance ratio. The results show that approximately 90% of the total variance can be explained by the first 20 principal components. This indicates that a substantial part of the variability in the data can be represented in a lower-dimensional space, which is consistent with the correlated structures observed in the correlation matrix.

Figure 1.3(b) shows the observations projected onto the first two principal components and coloured according to customer revenue. No clearly separated clusters are visible, suggesting that customers do not naturally form distinct groups based on the available features. Instead, the observations form a continuous cloud, with substantial overlap between low- and high-revenue customers.

Overall, the PCA confirms that the dataset contains correlated and partially redundant information, but also shows that customer revenue cannot be separated easily using only the first two principal components. For this reason, PCA was kept as an exploratory analysis and was not used as a preprocessing step in the final modelling pipeline.

1.2.3 Data Cleaning

Unused Features

After a first analysis of the raw dataset some variables did not seem relevant for the future model. Those features were only gave information about details like the printed patterns of the shoes or the size of the heels. The amount of missing values for those features was also substantial and it has been decided to drop them. The features in question are `prod_clasp`, `prod_comfort_wear`, `prod_comfort_sole`, `prod_print`, `prod_insole`, `prod_heel`, `prod_type_4`.

Missing Values

An evaluation of the percentage of missing values for the remaining features was done and it appeared that for some, missing values issues needed to be addressed. For `prod_material` and `prod_type_5` the percentage of missing values being below 7% of the total amount of data, they were filled with the value *missing* directly in the existing columns. For `returned_to_shop_id`, 81.43% of the values are missing ones. However this feature has been considered essential for the analysis and could not be dropped. The missing values were treated the same way knowing that only 20% of the feature is made of pertinent data.

Duplicates and format inconsistency

An appraisal of the presence of duplicate data was done and less than 1% duplicated rows were found. Nevertheless the consistency of features involving many different labels was done too. For `prod_brand` some inconsistencies were spotted. After searching on the web several brand names appeared to be subunits of other ones. They were changed to the global brand name to have a common format. Bana & Co, Brako Anatomics, Gabor Kids and Ragazzi Sport have been respectively gathered with the brands Bana, Brako, Gabor and Ragazzi.

Additional Data Cleaning

To prepare the product attributes for feature engineering, several cleaning steps were applied. First, the variables `prod_material` and `prod_type_3` often contained multiple comma-separated categories; therefore, new variables were created that retain only the first (main) category to improve consistency. Second, the complex `prod_season` codes

were decomposed into three interpretable components: a season code (e.g., W, Z, MZ), a corresponding season label (Winter, Summer, Mid-season, etc.), and the collection year extracted from the final two digits. Additionally, a `season_type` variable was created to distinguish between main, mid-season, special, NOS, and consignment collections. These transformations simplify the original categorical fields and make them more suitable for modelling and analysis.

Outliers

To visualize outliers with the quantitative variables (`sale_revenue`, `sale_discount_applied`, `prod_size`), boxplots were used. An estimation of the percentage of data it represents was also given by using quantiles. In all three cases the outliers represents 10% or less of the data, so it was decided to keep them. For `prod_size`, 3 bins (kid size, standard, big size) were created to handle the issue. For the two remaining variables quantiles were used all the values.

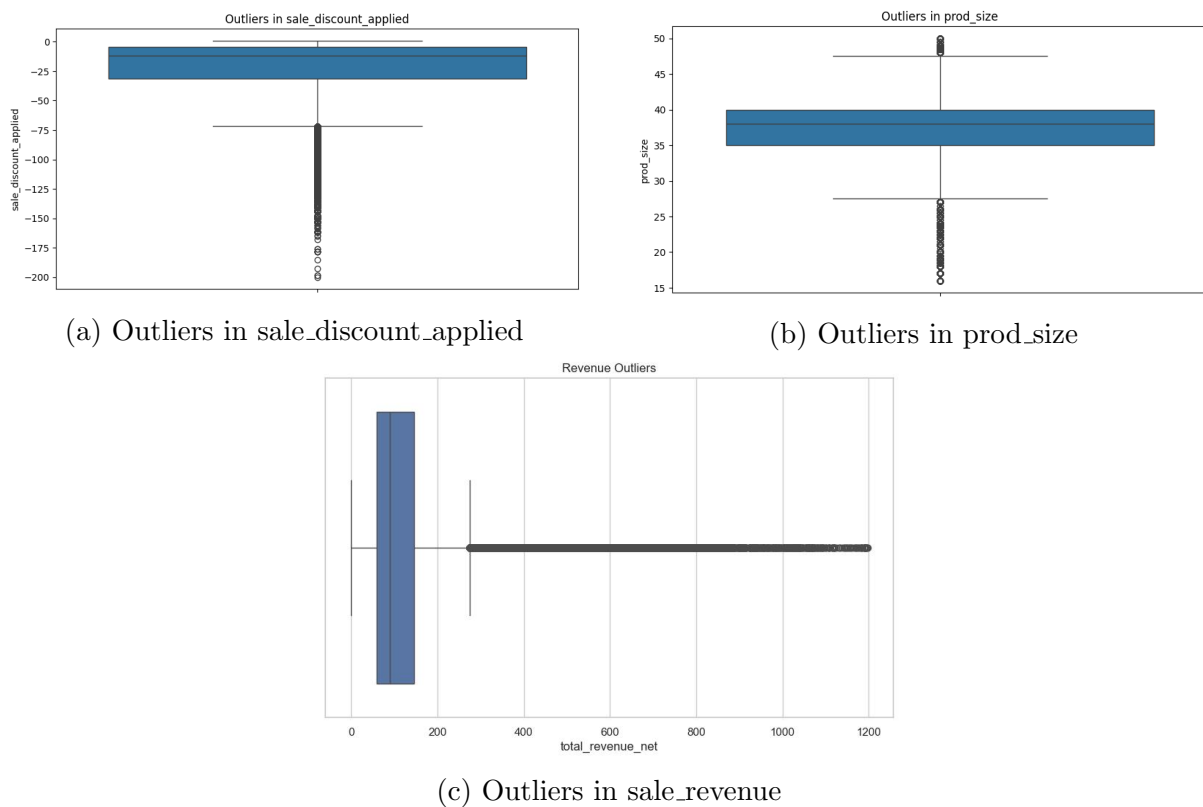


Figure 1.4: Boxplots of the quantitative variables.

1.3 Feature Engineering

To transform the raw transactions (2016–2017) into a high-dimensional customer-level matrix features were implemented. The engineered features capture behavioral footprints across two distinct methodologies: empirical transactional aggregations and advanced stochastic lifetime modeling.

The feature space can be structured into six major functional dimensions:

- **Core RFM and Lifetime Metrics:** Basic baseline indicators capturing the total volume of transactions (`n_sales`), temporal boundaries of the customer relationship (`customer_lifetime_days`), and density ratios evaluating how clustered a customer’s history is relative to their first purchase window (`recency_ratio`).
- **Purchase Intervals and Inter-purchase Gaps:** High-resolution features calculating the average, minimum, and maximum days between consecutive orders. Metrics such as `recent_gap_trend` track whether a customer’s purchasing frequency is expanding or contracting over time.
- **Recency Windows and Velocity:** Rolling-window metrics computing activity slices over the last 30, 90, 180, and 365 days. A key feature, `revenue_velocity`, isolates a client’s short-term daily expenditure (past 90 days) against their long-term average to capture accelerating or cooling purchase interest.
- **Operational and Diversity Preferences:** Variables analyzing explicit basket attributes, including brand and color loyalty metrics (`top_brand_share`), sizing profiles, and shoe size standard deviations to identify single vs. multi-consumer accounts.
- **Discount, Outlet, and Return Leakage:** Financial adjustments tracking discount sensitivity (`avg_sale_discount_ratio`) and product returns (`return_rate`, `returned_revenue_sum`), which act as crucial negative constraints for bottom-line revenue calibration.

1.3.1 Stochastic Lifetimes Enrichment

While empirical aggregations provide an excellent historical reflection, they are inherently backward-looking. To equip the downstream machine learning models with forward-looking probabilistic priors, the framework is enriched using the specialized `lifetimes` library. This layer executes two parametric models:

1. The BG/NBD Model (Beta-Geometric/Negative Binomial Distribution):

This stochastic model isolates purchase frequency and churn risk simultaneously. It models a customer's purchase cadence as a continuous Poisson process that stays active until the client abruptly "dies" (permanently churns). This yields two invaluable features:

- **prob_alive:** The conditional probability (0 to 1) that a customer is still actively retained on the cutoff date, mitigating the linear blind spots of basic recency metrics.
- **pred_num_purchases:** A pure predictive projection estimating the exact number of unique transactions the customer will complete in the upcoming 24-month target period.

2. **The Gamma-Gamma Model:** Because purchase frequencies and monetary values are assumed to operate independently, a Gamma-Gamma sub-model is layered on top of the active repeat-buyers. It functions as a Bayesian smoother, deriving **expected_avg_sales**. This calculates the expected average order value for future transactions, effectively shrinking extreme, isolated financial outliers back toward a realistic population mean.

1.4 Model Selection

1.4.1 Model Performance Evaluation

Mean Absolute Error

The Mean Absolute Error (MAE) measures the average distance between the predicted and the true value:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

with n equal to the amount of observations or customers in our case.

The MAE can in other words tell how big of an error we can expect from the forecast on average. The lower the MAE value, the better the model predicts. An MAE value equal to 0 indicates no error in the forecast. When comparing multiple models, the model with the lowest MAE is considered better.

Spearman Correlation

Unlike the Mean Absolute Error (MAE), which evaluates the model’s ability to predict exact revenue amounts, the Spearman rank correlation coefficient assesses how effectively the model ranks customers. This ranking approach is particularly valuable for identifying top-tier customer segments, such as the top 10% highest-value customers.

The Spearman correlation coefficient (ρ) is calculated using the following formula:

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

where d_i represents the difference between the true and predicted ranks for each observation, and n denotes the total number of observations (customers).

The coefficient ranges from -1 to 1 . A correlation of 1 indicates a perfectly predicted customer ranking, whereas a value of 0 suggests the model’s predictions are equivalent to random guessing. Ultimately, the objective is to maximize this positive correlation.

1.4.2 Two-Stage Model Framework

As the revenue variable is both highly skewed and zero-inflated (63% of customers generate no revenue) (see Figure 1.1a), a standard one-step regression model would struggle to learn meaningful patterns. To address this, we implemented a two-stage modeling framework that separates the prediction of whether a customer will buy from how much they will spend. The mechanics of the implemented two-stage framework operate as follows:

1. **Stage 1 (Classification):** A probabilistic classifier is trained to estimate the probability $P(\text{buy}_i)$ that customer i will generate non-zero revenue during the target period (2018–2019).
2. **Stage 2 (Regression):** Concurrently, a regressor is trained exclusively on the subset of active customers (where true revenue > 0). To stabilize variance and minimize the impact of extreme outliers, the continuous target variable is log-transformed using $\log(x + 1)$.
3. **Final Inferred Revenue:** The final expected revenue $E(\text{Revenue}_i)$ for each customer is computed by multiplying the probability from Stage 1 with the back-transformed expected value from Stage 2:

$$E(\text{Revenue}_i) = P(\text{buy}_i) \times (\exp(\hat{y}_{\log, i}) - 1)$$

To find the optimal baseline pair before exploring deep learning and automated machine learning architectures, we systematically benchmark a diverse matrix of algorithms within this two-stage pipeline. The evaluated configurations are divided into four distinct algorithmic families:

- **Linear and Kernel Methods:** Combinations such as Logistic Regression paired with a Gradient Boosting Regressor, and Support Vector Classifiers (SVC) paired with Support Vector Regression (SVR). These models test the baseline linear and non-linear boundaries.
- **Tree-Based Ensembles:** Random Forest (RF) and Extra Trees (XT) pairs. These models are inherently robust to tabular noise and are leveraged to capture complex interactions among the engineered RFM and Lifetimes features.
- **Boosting Algorithms:** Gradient Boosting (GBR) and AdaBoost pairs. These iterative learners focus sequentially on hard-to-classify customer segments.
- **Distance-Based Models:** K -Nearest Neighbors (KNN), providing a non-parametric benchmark based on customer similarity in the feature space.

Every model configuration is consistently evaluated on an independent validation set (20% stratified split). The evaluation relies on two distinct metrics, as described in Section 1.4.1 to capture both magnitude accuracy and ranking performance: Mean Absolute Error (MAE) for absolute monetary deviations, and the Spearman rank correlation coefficient (ρ) to assess the quality of the customer prioritization list. Based on these metrics, the model using a GBR classifier and GBR regressor performed best (see Figure 1.5 and 1.6). However, as these tabular models ignore the chronological order of transactions, in the next section we introduce an LSTM to model the residuals.

Support Vector Classifiers (SVC) and Support Vector Regressors (SVR) were initially tested, but will not be included in the final LSTM comparison for two reasons. First, both models required substantially higher computational resources than the other baseline models, especially when using nonlinear kernels. Second, their predictive performance was consistently weaker, with higher MAE values and lower Pearson correlations as can be seen on Figure 1.5 and 1.6. This suggests that the nonlinear patterns these models are designed to capture are not strongly present in the data. In addition, SVM-based models generally do not scale well to medium-to-large tabular datasets, as their training time grows rapidly with the number of samples and features. For these reasons, SVC and SVR were not retained for further comparison in the LSTM section.

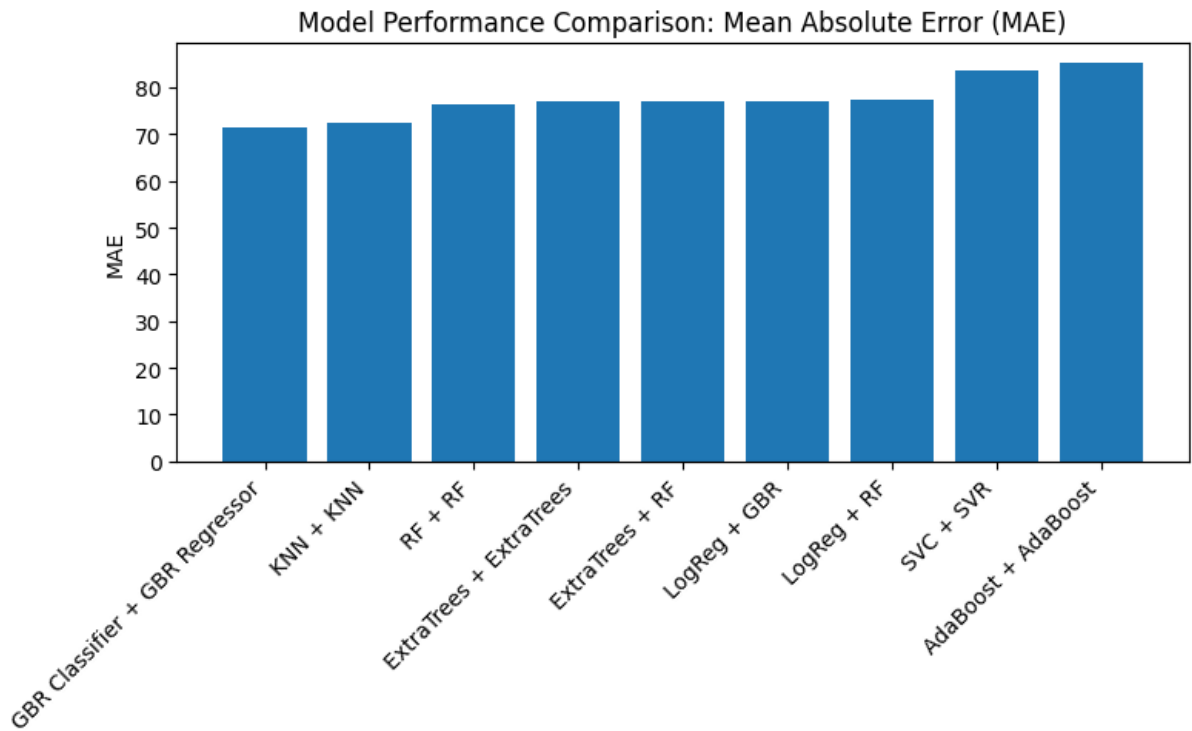


Figure 1.5: Model performance: Mean Absolute Error

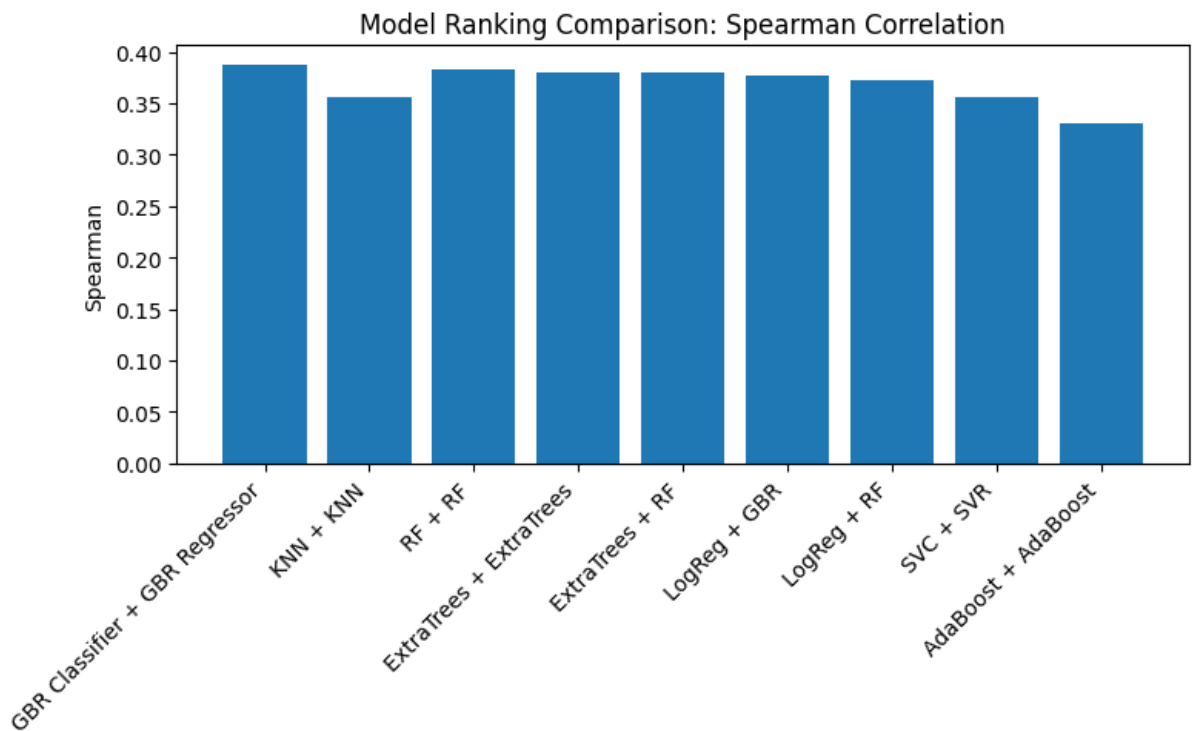


Figure 1.6: Model performance comparison: Spearman Correlation

1.4.3 Temporal Dynamics and Residual Modeling via LSTM

While traditional tabular machine learning models (such as Random Forests or Gradient Boosting) excel at processing static, aggregated data features, they are inherently blind to chronological sequence and temporal momentum. In customer transaction histories, the timing of purchases contains critical behavioral signals. To capture these underlying sequential patterns, we introduce a Long Short-Term Memory (LSTM) neural network. An LSTM is a specialized architecture of Recurrent Neural Networks (RNNs) designed to process sequence data without suffering from the vanishing gradient problem.

In our hybrid architecture, the LSTM is deployed as a sequential error-correction mechanism. Instead of predicting the final revenue directly, the LSTM is trained on the log-residuals (the prediction errors) of the two-stage baseline models, utilizing a 3D tensor of the 24-month chronological purchase timelines from 2016 to 2017. The network identifies remaining temporal trajectories that the baseline models missed, effectively fine-tuning the static predictions.

The experimental results, see Figure 1.7 and 1.8, demonstrate a profound architectural breakthrough, particularly regarding the Mean Absolute Error (MAE). The baseline models, operating purely on static tabular data, achieved baseline MAEs ranging between €71.18 and €77.13. However, upon integrating the hybrid Residual LSTM, every single model configuration experienced a massive performance surge. The top-performing baseline pair, GBR Classifier & GBR Regressor, saw its MAE drop from €71.18 to €64.82. Tree-based ensembles (such as RF & RF and ExtraTrees & RF) saw their errors slashed by over €11.00, converging closely around the €65.00 mark.

This uniform, systematic drop in MAE proves that the baseline models consistently miscalculated the monetary value of specific customer segments due to their inability to digest the purchase velocity. The LSTM successfully extracted this velocity from the 24-month transaction streams, correcting the magnitude of the errors in the log-space before back-transformation.

In contrast to the dramatic shifts in MAE, the Spearman rank correlation (ρ) remained exceptionally stable across all architectures, hovering tightly between 0.3770 and 0.3886. This divergence highlights an important distinction between absolute prediction accuracy and ranking performance. While MAE is highly sensitive to the numerical calibration and scale of the predicted CLV values, Spearman's ρ only evaluates whether customers are ranked in approximately the correct order. As a result, the LSTM can substantially alter the magnitude of the predictions, and therefore produce large changes in MAE, without materially changing the relative ordering of customers.

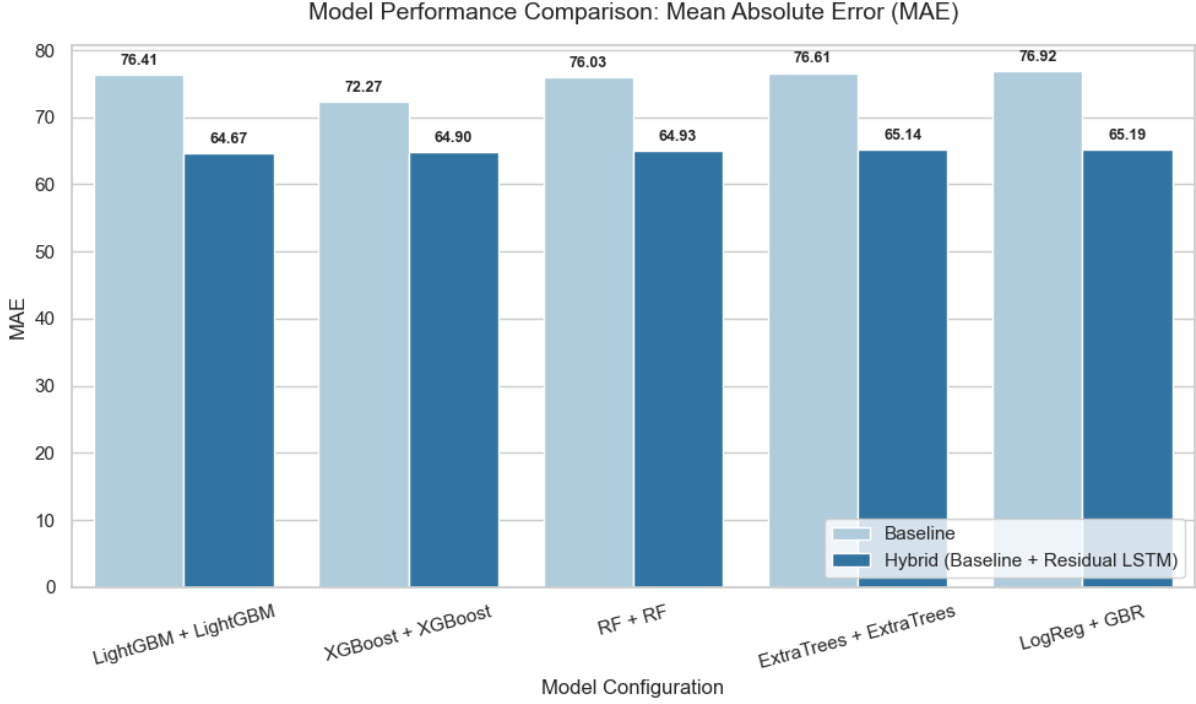


Figure 1.7: Model Performance Comparison: MAE before and after LSTM

The GBR Classifier & GBR Regressor augmented with the Residual LSTM emerged as the absolute champion, achieving the highest ranking accuracy of $\rho = 0.3886$. Across nearly all configurations (e.g., GBR, ExtraTrees, KNN), the hybrid LSTM approach marginally but consistently outperformed or matched the tabular baseline. This behavior indicates that the baseline models were already relatively competent at sorting customers hierarchically (identifying high-value vs. low-value clients) using aggregated RFM and lifetimes features.

Ultimately, the hybrid GBR & Residual LSTM framework marks the peak of manual architectural engineering, yielding a highly optimized combination of low absolute error (€64.82) and superior prioritization capability ($\rho = 0.3886$).

Although the hybrid LSTM approach improved the ranking, manual architecture engineering has its limitations. We therefore conclude by introducing AutoGluon to test the absolute performance limit through multi-layer stacking and automated ensembling.

1.4.4 Automated Machine Learning: AutoGluon Framework

To test the absolute performance limits of our feature-engineered dataset, we transition from manual architectural configurations to Automated Machine Learning (AutoML) via the AutoGluon-Tabular framework. While manual two-stage configurations and deep learning networks require arbitrary heuristic choices regarding model architectures and

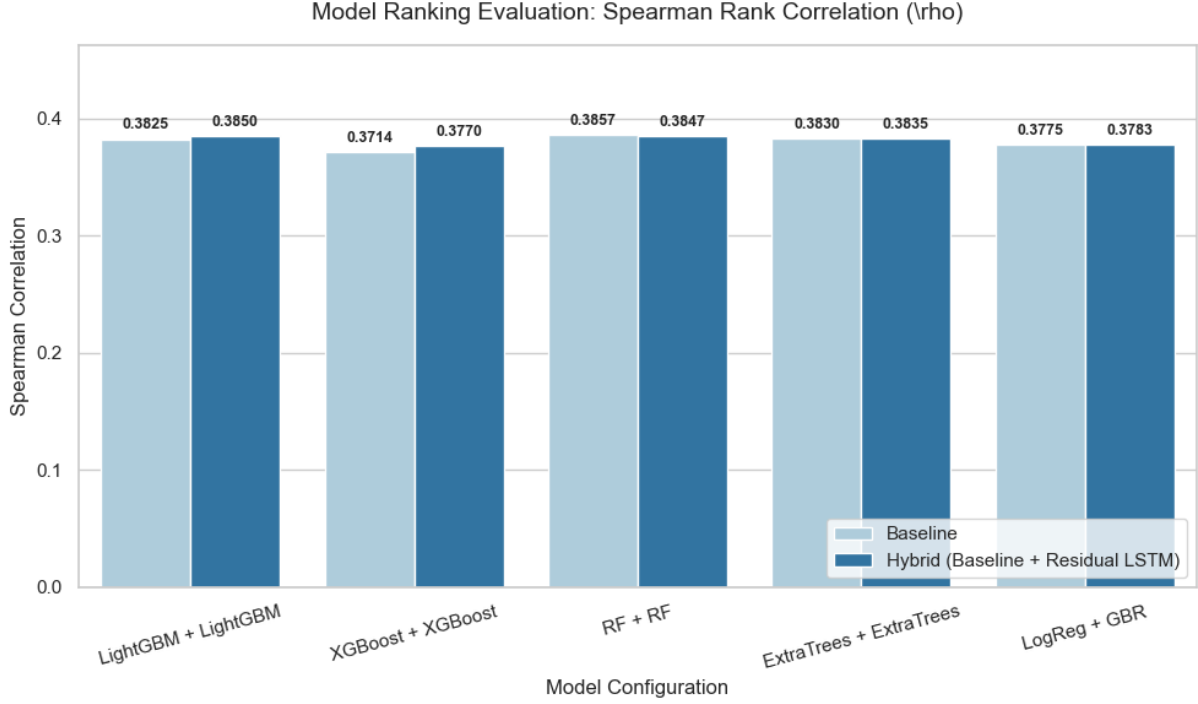


Figure 1.8: Model Performance Comparison: Correlation before and after LSTM

error-correction loops, AutoGluon bypasses these constraints. It leverages an advanced multi-layer stacking and bagging strategy, combining diverse algorithmic families into a highly optimized, weighted meta-ensemble.

The Impact of Target Log-Transformation on AutoML Performance

To evaluate whether mitigating the extreme right-skewness of the Customer Lifetime Value (CLV) distribution could further unleash the predictive power of Automated Machine Learning, we benchmarked the standard AutoGluon ensemble against an identical AutoGluon pipeline trained on the log-transformed target variable $\log(\text{revenue} + 1)$. The empirical validation results, illustrated in the Figure 1.9, demonstrate an improvement in both magnitude evaluation (MAE) and customer prioritization accuracy (Spearman's ρ)

By transitioning to the log-space, MAE decreased from €61.5855 to €60.4000. While a raw financial target forces regression algorithms to aggressively adjust decision splits to accommodate extreme high-value outliers, the log-transformation compresses this monetary variance. Consequently, the loss functions of the underlying base learners were no longer heavily distorted by extreme right-tail variance. This allowed AutoGluon to achieve a more mathematically stable calibration across the entire distribution of the customer base, directly translating into a tighter, more accurate absolute error boundary when back-transformed via $\exp(x) - 1$. The consistent reduction in MAE across all baseline models suggests that chronological purchasing patterns contain predictive information

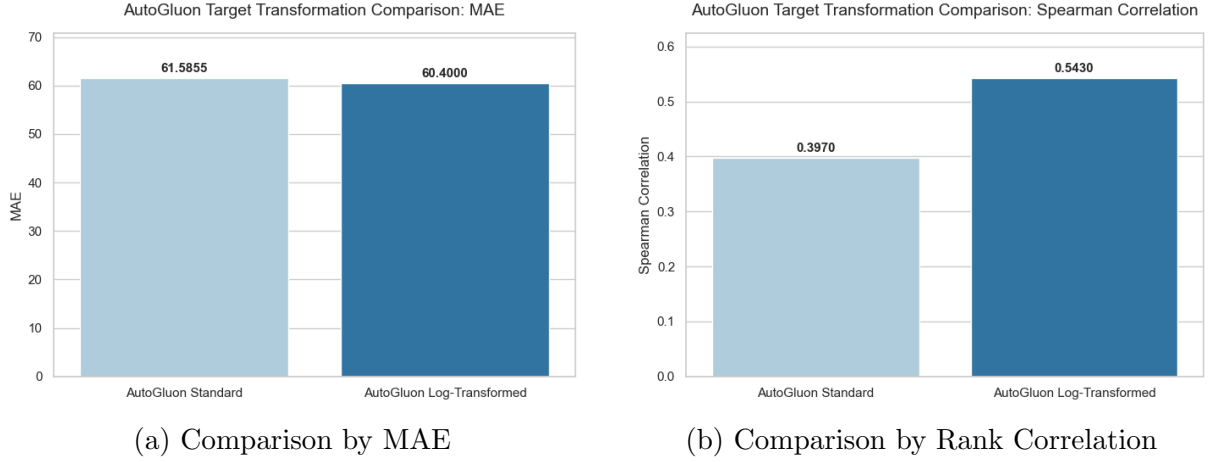


Figure 1.9: AutoGluon Performance Comparison regarding transformation target variable

beyond that captured by the engineered tabular features. This indicates that customer behaviour is influenced not only by historical aggregates but also by the temporal ordering and evolution of transactions.

The most remarkable empirical revelation is the massive surge in the Spearman rank correlation coefficient, which skyrocketed from 0.3970 to an unprecedented 0.5430. A leap of this magnitude implies that the target transformation fundamentally altered the internal ensemble composition and behavior of the AutoML framework. There are two primary structural explanations for this massive optimization: Firstly, are the neural networks embedded within the AutoGluon framework highly sensitive to unscaled, high-variance targets. In the raw Euro-space, these neural layers often fail to converge or yield erratic gradients, rendering their predictions useless within the final meta-ensemble. In the log-space, however, the target distribution approaches normality. This completely unlocks the deep learning components of AutoGluon, enabling them to discover complex, non-linear feature interactions that gradient-boosted trees might overlook.

Secondly, does the AutoGluon model use multi-layer stacking, where predictions of base models are fed as features into subsequent meta-learners. In a heavily skewed target environment, subtle ordering errors in earlier layers scale additively into massive variance penalties in later stages. The log-transformation enforces a smoother, monotonically stable space. This allows the final ensemble weighting layer to synchronize the models much more effectively, generating a highly accurate hierarchical sorting of customers.

Ensemble Weights and Architectural Decomposition

An architectural decomposition of the final meta-ensembles provides empirical proof of why the log-transformed framework achieved its predictive breakthrough. Under the

raw Euro-space framework (AutoGluon Standard), the pipeline converged at a standard two-layer stacking architecture (`WeightedEnsemble_L2`), combining two structural variants of CatBoost at the primary layer (`CatBoost_r6_BAG_L1` with a 76.00% weight, and `CatBoost_r167_BAG_L1` with a 24.00% weight).

Conversely, mitigating the extreme right-skewness through the log-transformation successfully unlocked a deeper level of ensembling logic, prompting AutoGluon to deploy multi-layer stacking (`WeightedEnsemble_L3`). In this stabilized environment, the predictions of the primary models were iteratively fed as features into a third layer of advanced meta-learners. The final prediction was mathematically driven by a new combination of second-layer models, namely `CatBoost_r5_BAG_L2` (75.00%) and `CatBoost_r6_BAG_L2` (25.00%).

As structured in Table 1.2, CatBoost variants commanded 100% of the active ensemble weights across both experimental frameworks. This absolute hegemony underscores that symmetric tree structures and robust categorical handling are mathematically superior for this noisy Customer Lifetime Value dataset. More importantly, it demonstrates that the multi-layer stacking enabled by the log-space serves as the primary catalyst for the massive surge in ranking accuracy ($\rho = 0.5430$).

Table 1.2: Active Ensemble Weights for AutoGluon Architectures

Architecture	Ensemble Level	Base Learner	Weight
Standard	WeightedEnsemble_L2	CatBoost_r6_BAG_L1	0.76
		CatBoost_r167_BAG_L1	0.24
Log-Transformed	WeightedEnsemble_L3	CatBoost_r5_BAG_L2	0.75
		CatBoost_r6_BAG_L2	0.25

Feature Importance

To isolate the predictive drivers within the final machine learning frameworks and improve model interpretability, a permutation importance analysis was conducted for both the standard and log-transformed AutoGluon models. In addition, feature importance was inspected for the manually engineered two-stage Gradient Boosting framework, consisting of a GBR Classifier and a GBR Regressor. This comparison is useful because the two modelling strategies differ fundamentally in how they exploit customer information: the manual approach imposes a predefined modelling structure, while AutoGluon automatically searches across model classes, transformations, and ensemble structures.

The manually engineered two-stage model (see Figure 1.10) displays a clear separation of roles between classification and regression. In the GBR Classifier, `recency_ratio` dom-

inates the feature-importance profile by a substantial margin. This indicates that the classification stage is primarily driven by customer activity and churn-related timing signals when distinguishing customers with high versus low future purchase potential. Other variables, such as `prod_type_5_n_unique`, `size_min`, `pred_num_purchases`, `prob_alive`, and `n_sales`, contribute only marginally in comparison. The classifier therefore functions mainly as a behavioural activity filter, relying heavily on whether a customer still appears commercially active.

The GBR Regressor, in contrast, is dominated by monetary information. The most important feature is `total_revenue_net`, followed by `revenue_last_365d` and `pred_num_purchases`. This suggests that once the model has identified customers likely to generate future value, the regression stage mainly uses historical revenue magnitude to determine the expected revenue. In other words, the manual two-stage architecture separates the problem into two relatively interpretable components: a recency-driven classification step that identifies customer viability, and a revenue-driven regression step that scales the predicted customer value.

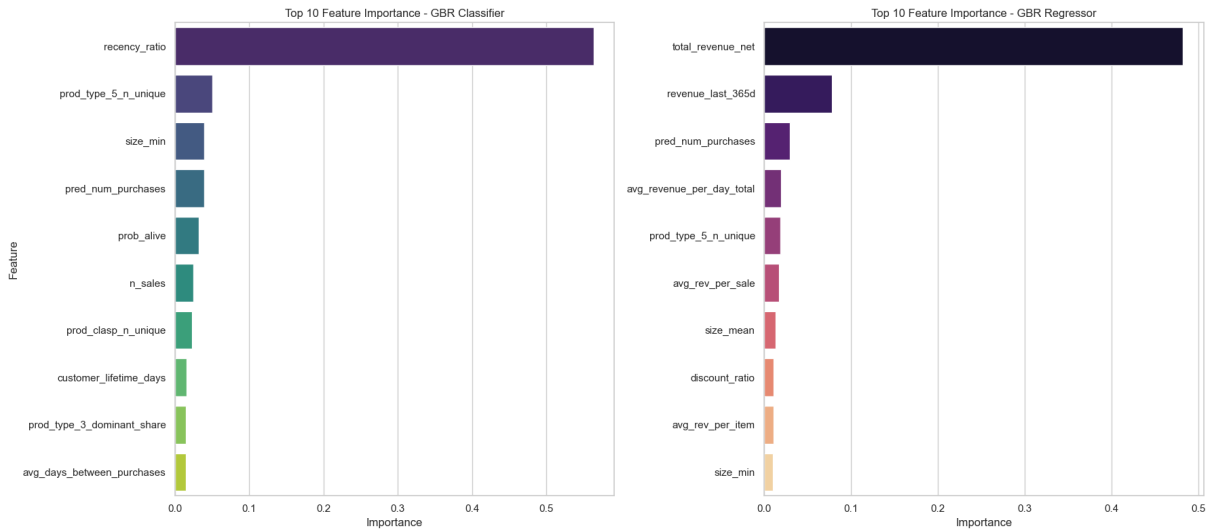


Figure 1.10: Feature Importance GBR Classifier and Regressor

The AutoGluon models (see Figure 1.11 a broader and more integrated use of the available feature space. In the standard AutoGluon model, the most important predictors are `avg_days_between_purchases`, `recency_ratio`, `n_sales`, `purchase_frequency`, and `prob_alive`. This indicates that AutoGluon does not rely exclusively on monetary aggregates, but combines purchase timing, frequency, customer lifetime, and survival-style features. Nevertheless, monetary information remains relevant in the standard model, with `total_revenue_net` still appearing among the top-ranked predictors. This is consistent with the untransformed CLV target: because large-spending customers can generate substantial absolute errors, the model must still anchor its predictions to historical revenue magnitude.

The log-transformed AutoGluon model shows a more pronounced shift toward behavioural and lifecycle signals. Here, `recency_ratio` becomes the dominant feature, followed by `prob_alive`, `n_sales`, `customer_lifetime_days`, and `pred_num_purchases`. Raw monetary variables become less central, with `revenue_last_365d` only appearing in the lower part of the top ten and `total_revenue_net` falling outside the most influential features. This shift is a direct consequence of the log-transformation: by compressing extreme CLV values, the model becomes less dependent on absolute revenue scale and more focused on signals that describe customer health, purchase regularity, and relationship maturity.

This comparison clarifies the difference between the manual and automatic modelling approaches. The manually engineered GBR framework relies on an explicitly designed decomposition of the CLV problem: classification is mainly governed by recency and activity indicators, while regression is mainly governed by monetary history. AutoGluon, by contrast, learns a more blended representation in which timing, frequency, lifetime, survival probability, and monetary features are evaluated jointly. Especially in the log-transformed setting, AutoGluon appears to exploit behavioural structure more effectively, rather than simply extrapolating from past revenue.

Therefore, the feature-importance results provide a plausible explanation for the performance improvement observed with AutoGluon. The manual two-stage model captures the main intuitive CLV mechanisms, namely customer activity and historical spending, but it does so through a constrained architecture. AutoGluon is able to search over more flexible ensembles and automatically combine multiple complementary signals. As a result, it can exploit additional interactions between recency, frequency, lifetime, purchase regularity, and survival-based features that are only partially captured by the manually engineered pipeline. This broader use of the feature space likely contributes to its stronger predictive performance.

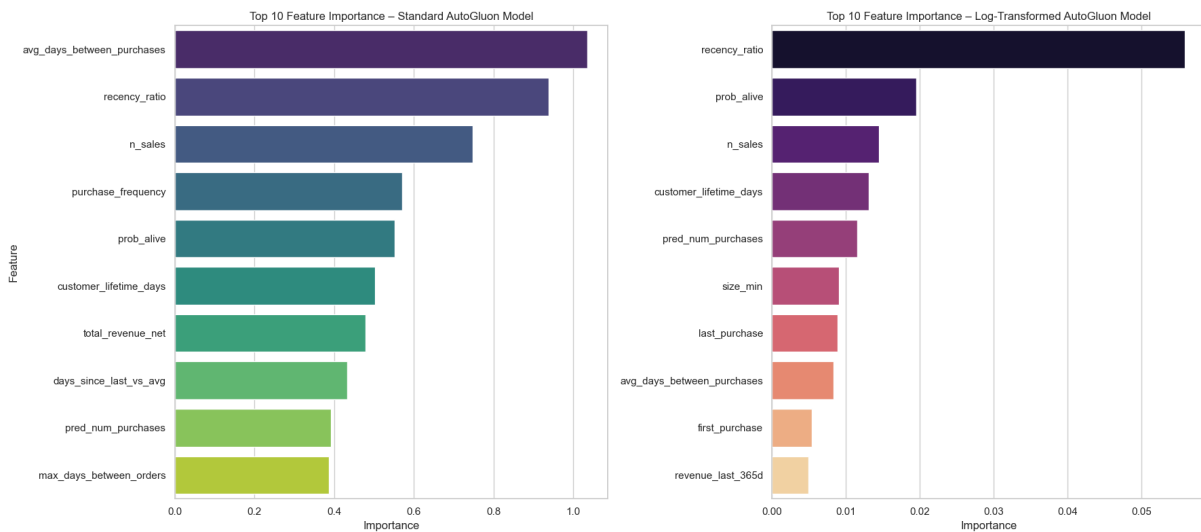


Figure 1.11: Feature Importance AutoGluon Models

1.5 Conclusion

Ultimately, the AutoGluon Log-Transformed architecture stands as the strongest model in this research. By combining advanced feature engineering with automated multi-layer ensembling in a stabilized mathematical target space, it successfully addresses both strict financial calibration (€60.40 MAE) and high-quality customer prioritization (Spearman $\rho = 0.5430$). This configuration is therefore selected for final test-set deployment.

For the public leaderboard submission, however, only the standard AutoGluon model could be submitted in time, where it ranked 19th. Although the log-transformed variant was not included before the deadline, its superior validation performance suggests that it would likely have achieved a substantially higher position.

Chapter 2

Assignment 2: Deep learning on images

2.1 Introduction

This project aims to classify web-scraped LEGO images by integrating visual information extracted from the images with the accompanying metadata. The overall objective is to examine how different modeling strategies, including image-based and metadata-enhanced approaches, can be leveraged to achieve accurate category-level classification of LEGO items.

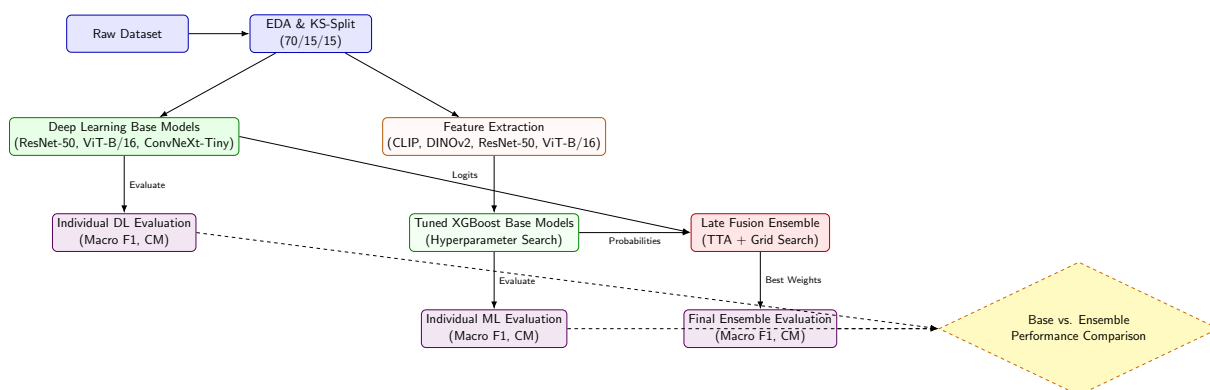


Figure 2.1: Assignment 2 image processing workflow.

The analysis began with data exploration and data cleaning to better understand the structure, quality, and consistency of both the scraped images and the associated metadata. After this initial preparation, the dataset was split on a per-category basis into training, validation, and test sets using a ratio of 70%, 15%, and 15%, respectively. This partitioning strategy ensured that each category was represented across all subsets while supporting a rigorous deep learning workflow. More specifically, the validation set was

used for model selection and early stopping, whereas the test set was reserved for the final assessment of model performance.

Several modeling strategies were then explored. First, a set of pretrained deep learning architectures was employed and subsequently fine-tuned for the image classification task. The fine-tuned models included Vision Transformer (ViT-B/16), ResNet-50, and ConvNeXt-Tiny, allowing comparison across transformer-based and convolution-based architectures with different levels of representational complexity. In addition, a hybrid approach was developed in which image embeddings were extracted and combined with informative metadata features. For this purpose, embeddings were generated using CLIP, DINOv2, ResNet-50, and ViT-B/16. These joint feature representations were then used as input for a tuned XGBoost model. Finally, test-time augmentation (TTA) and ensemble strategies were applied to further improve predictive performance and enhance the robustness of the final classification results.

For all models, performance was evaluated using the macro F1 score. This metric was selected because it assigns equal importance to each class, making it particularly appropriate when class distributions are imbalanced and when balanced performance across all categories is of primary interest.

2.2 Data Exploration and Data Cleaning

This section examines the distribution of the dataset and the overlap between subcategories and themes. Since Lego Minifigs are organized according to several related metadata fields, such as category, subcategory, and themes, it is important to understand how these labels are distributed before constructing the prediction task. In particular, exploratory analysis helps reveal potential class imbalance, visually or semantically similar groups, and cases where the same figure may be associated with multiple themes. These patterns are relevant for later modeling decisions, including the choice of a top-k classification task, the construction of train, validation, and test splits, the use of macro-F1 as the main evaluation metric, and the design of early stopping criteria. Therefore, the exploratory analysis serves as the basis for defining a modeling workflow that is both practically feasible and appropriate for the structure of the data.

Full access to Assignment 2 materials and implementation details is available at <https://github.com/zhaoyuwang0606-rgb/Big-data-project-image-processing>.

2.2.1 Class Distribution

In this classification task, class distribution has a direct impact on selection of model evaluation metrics. When the data are severely imbalanced, macro F1 is a more appropriate metric than overall accuracy, particularly when performance on smaller classes is of substantive importance.

The original dataset exhibits a pronounced long-tail distribution across categories. As shown in Figure 2.2, a small number of categories account for a large share of the observations, whereas many categories contain only a very limited number of images. In this paper, the analysis was restricted to the 30 most frequent categories. This decision was made for two main reasons. First, very small categories do not contain enough observations to support a meaningful train, validation, and test split. Second, the computational cost of training and tuning deep learning models across the full set of categories would be disproportionate given the available resources. Focusing on the top 30 categories therefore provides a more reliable and computationally feasible classification setting while still covering the most substantively relevant portion of the dataset.

To address this imbalance, macro F1 was adopted as the primary evaluation metric. Unlike overall accuracy, macro F1 assigns equal weight to each category regardless of its size, making it more suitable when performance on minority classes is also of interest. In other words, the metric summarizes classification performance by first computing an F1 score for each category and then averaging these scores across all categories. Macro F1 is defined as follows:

$$Macro\ F1 = \frac{1}{K} \sum_{k=1}^K F1_k$$

where K denotes the number of categories and $F1_k$ represents the F1 score for category k . The class-specific F1 score is defined as:

$$F1_k = 2 \cdot \frac{Precision_k \cdot Recall_k}{Precision_k + Recall_k}$$

where $Precision_k$ and $Recall_k$ denote the precision and recall for category k , respectively.

Overall, the class distribution analysis highlights two central considerations for the present study: the need to restrict the task to sufficiently represented categories and the need to evaluate performance with a metric that is robust to imbalance. With these distributional properties established, the next subsection turns to the overlap between labels and the semantic similarity between categories, both of which introduce an additional source of

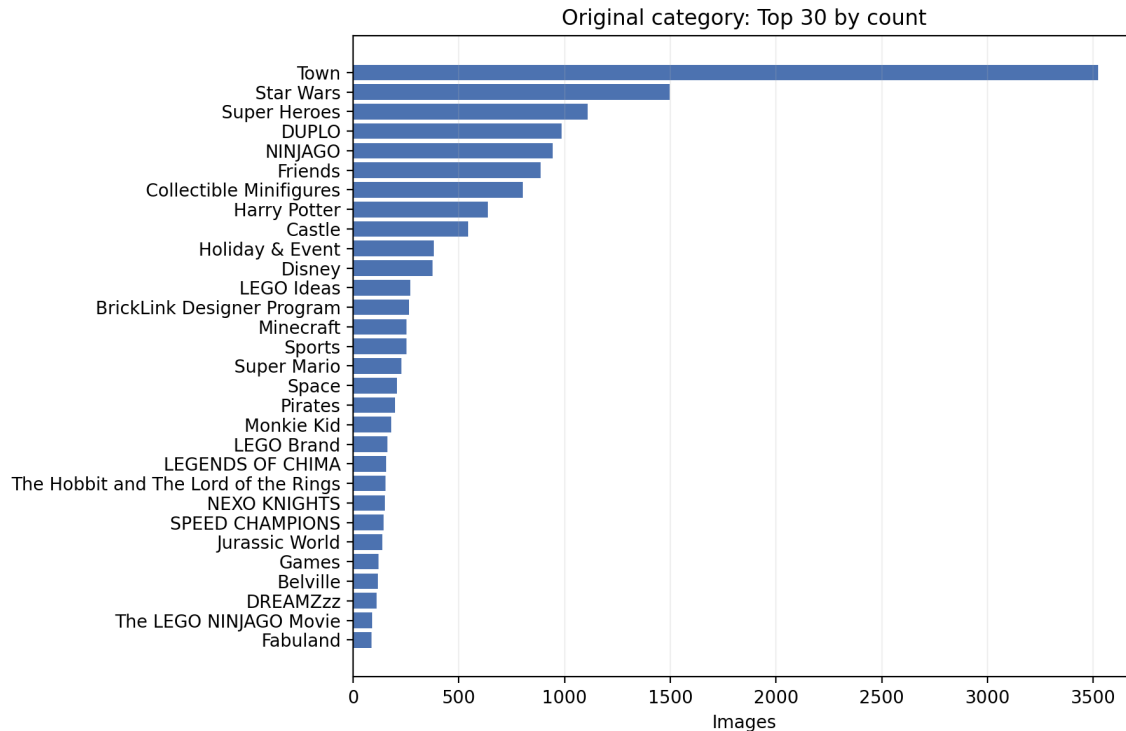


Figure 2.2: Category histogram of the LEGO Minifigs dataset.

classification difficulty.

2.2.2 Label Overlap and Semantic Similarity

A second important aspect of the exploratory analysis concerns the extent to which sub-categories and themes overlap. In the LEGO Minifigs dataset, some classes are closely related in meaning, visual appearance, or thematic context, which can make the classification task inherently ambiguous. Examining these overlaps helps identify where category boundaries are less distinct and where confusion between labels is more likely to occur. This is particularly relevant for interpreting model errors and for motivating the use of evaluation measures that reflect performance across all classes rather than only the most frequent ones.

The overlap structure is illustrated from two complementary perspectives. First, the sub-category overlap heatmap highlights where fine-grained labels share substantial similarity and are therefore more likely to be confused by the model. Second, the theme overlap visualization provides a broader view of how higher-level thematic groups intersect across the dataset. Together, these figures help explain why some misclassifications are structurally more difficult to avoid than others.

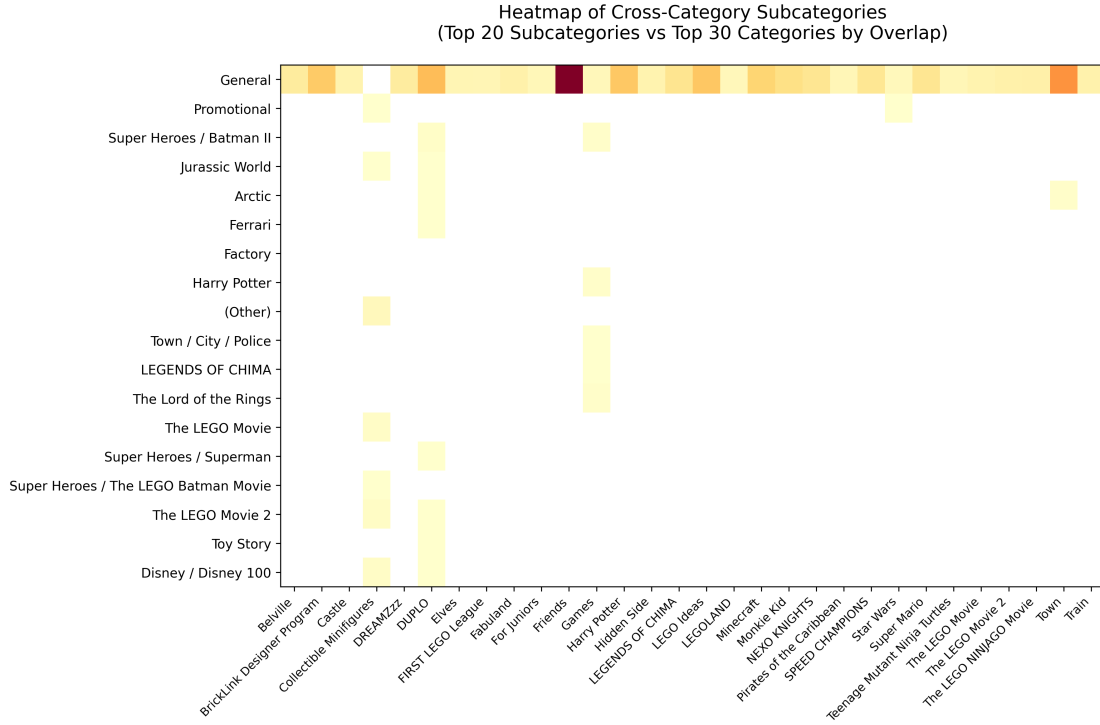


Figure 2.3: Subcategory overlap structure.

Beyond class imbalance, the dataset also exhibits substantial semantic ambiguity arising from its hierarchical and non-mutually exclusive metadata structure, particularly in the theme and subcategory labels. This ambiguity makes category boundaries less distinct and increases the likelihood of systematic misclassification.

A similar pattern emerges at the subcategory level. Figure 2.3 highlights extensive cross-category overlap in the most recurrent subcategory labels. The clearest example is *General*, which appears across 109 different primary categories, indicating that this label carries limited discriminative value on its own. Other subcategories, such as *Promotional* and *Jurassic World*, also occur across multiple categories, further demonstrating that the metadata structure introduces substantial label leakage and semantic overlap into the classification task.

Figure 2.4 provides a more nuanced view of theme-level ambiguity by showing both the percentage of minifigures associated with multiple themes and the corresponding raw counts. The figure indicates that *DUPLO* stands out most clearly on both dimensions, with by far the highest multi-theme percentage and the largest absolute number of multi-theme minifigures. Several other categories, including *Pirates* and *Castle*, also show comparatively high percentages of multi-theme assignment, suggesting that theme overlap is especially pronounced in these groups. By contrast, categories such as *Town* contribute a large raw number of ambiguous cases while exhibiting a much lower percentage overall, indicating that their ambiguity is driven more by dataset size than by an exceptionally

high relative overlap. Taken together, these patterns show that theme ambiguity arises in two distinct ways: through categories with high proportional overlap and through large categories that generate many ambiguous observations in absolute terms.

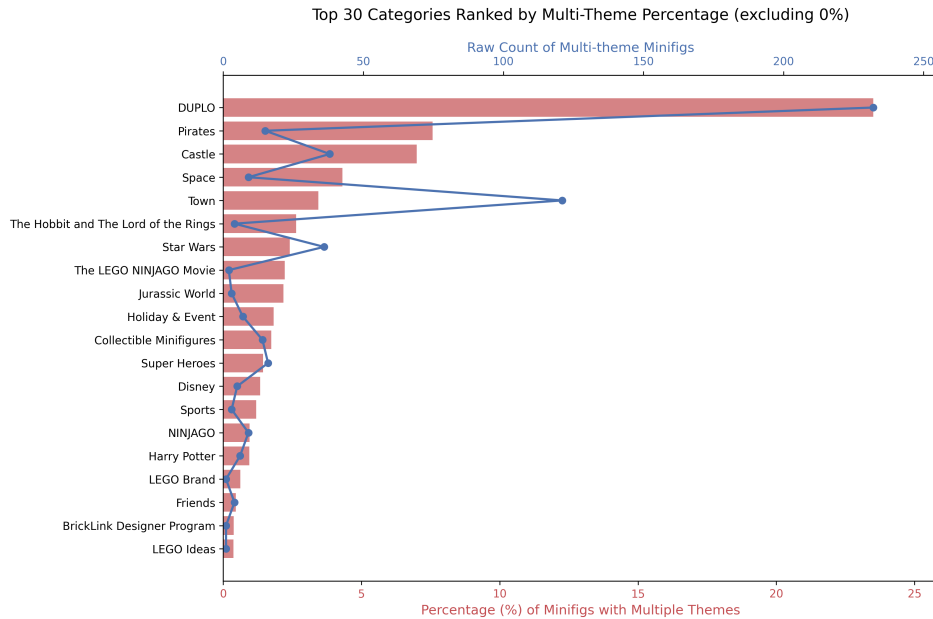


Figure 2.4: Theme overlap structure.

Overall, these findings suggest that the classification difficulty is not solely a consequence of class imbalance, but also of structural ambiguity embedded in the label system itself. This provides a clear motivation for exploring more sophisticated modeling strategies, since stronger models may be better able to capture subtle visual distinctions and contextual patterns. At the same time, the observed overlap also implies that performance gains from increased model complexity are likely to be bounded by the ambiguity of the labels themselves, meaning that not all errors should be interpreted as purely model-related.

2.3 Data Splitting

Given the pronounced class imbalance in the dataset, particular care was required when constructing the training, validation, and test partitions. A naive random split at the image level could easily lead to very small groups being concentrated in only one subset, which would make model training unstable and evaluation less reliable. To avoid this problem, the data were split on a group basis rather than purely at the individual-image level. This strategy helped ensure that smaller categories and related groups remained represented across the different subsets, thereby reducing the risk that an entire group would appear only in the training set or only in the validation set. As a result, the

final partitioning scheme provides a more balanced and methodologically sound basis for classifier development and performance assessment.

The 70/15/15 split was selected with the smallest retained category in mind. Because the least frequent category contains only 90 images, a more extreme split such as 80/10/10 would leave too few observations in the validation and test sets for reliable evaluation. In such a case, performance estimates would become highly sensitive to random variation, especially for the minority classes. Using 15% of the data for both validation and test therefore provided a more appropriate balance: it preserved the majority of the observations for training while still ensuring that even the smallest category remained sufficiently represented in the evaluation subsets. In addition, the split was performed in a stratified manner so that class proportions were maintained as consistently as possible across the training, validation, and test sets. To evaluate whether the training, validation, and test sets were sufficiently randomized, two validation steps were performed. First, we examined whether identical minifigures appeared across multiple subsets. This check indicated that the image files and corresponding metadata entries were unique in nearly all cases. The only exception concerned the minifigure *HOL183*, which appeared twice in the metadata but was linked to only a single image path. Consequently, only one instance entered the analysis pipeline, meaning that no effective duplication across subsets occurred.

Second, Kolmogorov–Smirnov tests were used as a split-quality diagnostic rather than as a feature-selection procedure¹. After stratifying by category, we wanted to verify that the training, validation, and test sets were also comparable with respect to non-leaking image-level metadata. This was important because a split can be balanced in terms of class frequencies while still being biased if, for example, newer minifigures or higher-value items are concentrated in one subset. Such a shift would make validation or test performance reflect a temporal or market-value distribution difference rather than genuine visual generalization.

For this reason, the KS checks focused on the variables that can characterize the image instances without directly revealing the target label: “year_released”, “current_value_new”, and “current_value_used”. Categorical metadata such as themes and subcategories were deliberately excluded from this diagnostic because they are semantically tied to the target categories and would introduce label leakage. The duplicate field “year” was inspected only as a redundancy check and was not treated as an additional source of information, since it contains the same information as “year_released”. Since the Kolmogorov–Smirnov test compares entire empirical distributions rather than only means or variances,

¹In simple terms, the Kolmogorov–Smirnov test compares two distributions by measuring the largest gap between their cumulative distribution curves. A larger gap indicates stronger evidence that the two samples may not come from the same underlying distribution. This makes the test useful here because it checks whether the train, validation, and test subsets have similar overall distributions, not only similar averages.

it provided an appropriate check for distributional differences in these numeric metadata variables across the three subsets. In total, the split diagnostic comprised 99 KS tests, combining the relevant pairwise subset comparisons with category-level checks for the non-leaking metadata variables. Only 3 of these tests rejected the null hypothesis of equal distributions. This corresponds to a rejection rate of approximately 3.0%, which is below the nominal 5.0% significance level. Therefore, the small number of rejections is consistent with ordinary random testing variation and does not indicate systematic distributional imbalance across the splits. Overall, these findings support the conclusion that the data split was sufficiently randomized, which strengthens the validity of validation-based early stopping and final test evaluation.

2.4 Model Selection and Training details

This section provides an overview of the models selected for training in this study. The final modeling pipeline included several fine-tuned deep learning architectures, namely ResNet-50, ViT-B/16, and ConvNeXt-Tiny, together with metadata-enhanced machine learning models based on image embeddings and tuned XGBoost classifiers. In addition, ensemble-based variants were considered in order to combine the strengths of different model families within the final prediction framework. Besides outlining the set of trained models, this section also summarizes the main parameter settings used during model development, including optimization-related choices and training configurations. Finally, it describes the criteria adopted for early stopping and model selection, which played an important role in controlling overfitting and ensuring that model performance was evaluated consistently across experiments.

2.4.1 Fine-Tuning Strategy

We first adopted a strategy based on fine-tuning pre-trained deep learning models. In this setting, the use of ResNet-50, ViT-B/16, and ConvNeXt-Tiny was intended to introduce architectural diversity into the modeling process. ResNet-50 is particularly effective at capturing localized visual cues, such as torso prints, facial details, and surface textures, whereas ViT-B/16 is better suited to modeling more global semantic relations and broader shape or color configurations through its self-attention mechanism. ConvNeXt-Tiny provides a modernized convolutional architecture that preserves the inductive biases of convolutional neural networks while incorporating several design principles from more recent vision models.

Rather than training these architectures from scratch, all models were initialized with

pre-trained weights and subsequently fine-tuned on the LEGO image classification task. This strategy allowed the models to build on general visual representations learned from large-scale datasets, while adapting their higher-level features to the more specific characteristics of LEGO minifigures.

Since the models were initialized from pretrained image classifiers, a freeze-then-unfreeze fine-tuning strategy was applied. Specifically, the pretrained backbone was frozen for the first five epochs, while only the newly added classification head was trained. This warm-up phase allowed the randomly initialized head to learn a stable mapping from the pretrained visual features to the LEGO minifigure categories before the full network was updated. After this stage, the backbone was unfrozen so that the model could further adapt its visual representation to LEGO-specific details such as torso prints, helmets, accessories, and color patterns.

At the data level, several augmentation techniques were used to increase the effective diversity of the training set. Random resized cropping encouraged the model to learn from both local and global visual cues rather than memorizing a fixed image framing. Horizontal flipping and small rotations improved robustness to minor changes in image orientation, while color jittering reduced dependence on specific brightness, contrast, saturation, and hue conditions. In addition, random erasing masked parts of the image during training, discouraging the model from relying too heavily on a single discriminative region. CutMix provided further regularization by replacing image patches with patches from other samples and mixing the corresponding labels, thereby forcing the model to make predictions from partial and mixed visual evidence.

The augmentation parameters were chosen to be deliberately conservative. The LEGO images are fine-grained: small differences in torso prints, helmets, accessories, and colour patterns can be category-relevant, so overly aggressive transformations could distort the signal that the classifier needs to learn. For this reason, the resized crop scale was limited to 0.7–1.0, rotations were restricted to $\pm 15^\circ$, and colour jitter was kept moderate. Horizontal flipping was used with probability 0.5 because left-right orientation is generally not part of the class definition, whereas random erasing was kept at 0.2 so that the model is encouraged to use multiple regions without making too many training images unrecognizable. CutMix was applied with probability 0.5 to regularize the decision boundary, but it was combined with otherwise mild geometric transformations to avoid excessive label noise.

Table 2.1: Anti-overfitting and robustness measures used during model training

Level	Measure	Setting	Level	Measure	Setting
Fine-tune	Freeze / unfreeze	--freeze-epochs 5	Model reg.	Head dropout	p=0.5
Data aug.	Resized crop	scale=(0.7,1.0)	Loss reg.	Label smoothing	label.smoothing=0.1
Data aug.	H. flip	p=0.5	Optimizer	AdamW decay	--weight-decay 0.05
Data aug.	Rotation	$\pm 15^\circ$	LR ctrl.	Cosine anneal.	--maximum epochs 500
Data aug.	Color jitter	brightness=0.2, contrast=0.2, saturation=0.2, hue=0.05	Selection	Best val. macro-F1	Validation macro-F1
Data reg.	Random Erasing	--random-erasing-prob 0.2	Early stop.	Smoothed val. macro-F1	--early-stop-patience 15
Data reg.	CutMix	--cutmix-prob 0.5	Checkpoint	Best checkpoint	best.pt

Further regularization was applied at the model, loss, and optimization levels, and the selected values reflect the same balance between flexibility and overfitting control. A dropout probability of 0.5 was used in the classification head because this is the part of the network most directly adapted to the relatively small LEGO dataset after pretraining. Label smoothing of 0.1 was used as a mild loss-level regularizer: it discourages overconfident predictions without removing the class signal. AdamW weight decay was set to 0.05 to penalize unnecessarily large parameter updates during fine-tuning, especially after the pretrained backbone was unfrozen. The maximum epoch count of 500 should therefore be interpreted as an upper bound rather than as the intended training length. Cosine annealing allows the optimizer to continue making smaller updates if validation performance keeps improving, while early stopping prevents the model from using the full training budget once the validation macro-F1 has plateaued.

Finally, model selection was based on validation performance rather than training accuracy. Because the task involved imbalanced category frequencies, validation macro F1 was used as the main selection criterion. Early stopping was based on a smoothed validation macro F1 trend with a patience of 15 epochs². This reduced the risk of continuing training after validation performance had stopped improving. Final test evaluation was conducted using the best validation checkpoint rather than the final epoch checkpoint.

2.4.2 Multi-modal Feature Fusion with XGBoost and Ensemble Strategy

In the previous subsection, we described a strategy in which pretrained image classifiers were directly fine-tuned while several regularization and overfitting-prevention techniques were applied during training. Although this approach allowed the models to learn strong visual representations from the LEGO images, it did not yet make explicit use of

²Let F_t denote the validation macro F1 obtained at epoch t . The 3-epoch simple moving average is defined as $SMA_t = \frac{1}{3}(F_t + F_{t-1} + F_{t-2})$. This smoothing was used because the raw macro F1 values tended to fluctuate substantially from epoch to epoch, especially under class imbalance and limited validation counts for minority classes. As a result, relying on the unsmoothed macro F1 could make early stopping overly sensitive to short-term noise rather than genuine performance improvements. The SMA therefore provided a more stable signal for model selection.

the available non-image metadata. This is a relevant limitation, since variables such as product-related attributes, release information, or other structured descriptors may contain complementary information that is not fully recoverable from the visual input alone. For this reason, a multi-modal strategy was also considered in which image-derived features were combined with metadata-based information in order to improve predictive performance. In particular, image embeddings extracted from the deep learning models were integrated with structured non-image features and used as input to tuned XGBoost classifiers. This setup made it possible to evaluate whether the combination of visual and tabular information could yield a more robust and discriminative representation than image-only fine-tuning.

The selection of CLIP, ResNet-50, ViT-B/16, and DINOv2 was not intended to suggest that these are the only suitable feature extractors or that they are universally optimal for this task. Rather, they were chosen as widely used and representative pretrained models that cover complementary families of visual representation learning. ResNet-50 represents the classical convolutional neural network paradigm and is particularly well suited to capturing local texture, edge, and pattern information. ViT-B/16 represents transformer-based visual modeling and is better suited to capturing more global relationships between image patches. CLIP contributes a vision-language pretraining signal, which can encode more semantic visual concepts, while DINOv2 provides strong self-supervised visual features with high transferability across downstream tasks. Taken together, these models yield diverse embeddings that allow the downstream XGBoost classifier to exploit different and potentially complementary aspects of the LEGO minifigure images.

XGBoost was selected as the downstream classifier because the fusion problem is tabular once image embeddings have been extracted. The input consists of fixed high-dimensional embedding vectors together with a small number of continuous metadata variables, rather than raw images. In this setting, gradient-boosted decision trees provide a strong and computationally efficient learner: they can model nonlinear interactions between embedding dimensions and metadata, require less feature scaling than distance-based or linear classifiers, and include built-in regularization through tree depth, learning rate, subsampling, column sampling, and boosting rounds. Simpler classifiers such as logistic regression would mainly test linear separability of the embeddings, while k-nearest neighbours and kernel SVMs are less attractive in high-dimensional embedding spaces because they rely heavily on distance geometry and can become computationally expensive. A small multilayer perceptron would be another possible choice, but it would introduce an additional neural model with more tuning sensitivity and a higher risk of overfitting on the limited validation data. XGBoost therefore offered a pragmatic compromise: it is expressive enough for nonlinear multimodal fusion, relatively robust on structured data, and produces class-probability outputs that can be incorporated into the final late-fusion

ensemble.

When integrating tabular metadata with visual features, careful feature selection was necessary to prevent data leakage. In particular, variables such as *subcategory* and *themes* were deliberately excluded. These categorical attributes are hierarchically and semantically coupled with the target labels, meaning that they can strongly constrain, and in some cases almost directly reveal, the correct class. For example, a theme label such as *DC Super Heroes* already narrows the set of plausible target categories substantially. Including such variables would therefore risk transforming the classification problem into a metadata lookup task rather than a genuine visual recognition problem, leading to overly optimistic performance estimates that would not reflect settings in which such metadata are unavailable at inference time. Instead, only continuous and non-deterministic contextual variables were retained, namely *year_released*, *current_value_new*, and *current_value_used*. The variable *year* was not included separately, because in this dataset it is identical to *year_released*³.

Table 2.2: Optuna search space for the metadata-augmented XGBoost classifier

Hyperparameter	Search Space
<i>max_depth</i>	Integer in [3, 10]
<i>learning_rate</i>	Log-uniform in [0.001, 0.3]
<i>subsample</i>	Float in [0.5, 1.0]
<i>colsample_bytree</i>	Float in [0.2, 1.0]
<i>gamma</i>	Log-uniform in [10^{-8} , 1.0]
<i>min_child_weight</i>	Integer in [1, 10]
<i>reg_alpha</i>	Log-uniform in [10^{-8} , 10.0]
<i>reg_lambda</i>	Log-uniform in [10^{-8} , 10.0]

These numerical features provide complementary prior information that can help the multimodal XGBoost classifier distinguish between visually similar but chronologically or economically distinct minifigure variants, without directly encoding the target label itself. For the metadata-augmented classifier, visual embeddings were concatenated with the selected numerical metadata variables and used as input to an XGBoost classifier. Hyperparameters were optimized using Optuna. In each trial, an XGBoost model was trained on the training set and evaluated on the validation set. The search objective was validation macro F1, which was selected because of the class-imbalanced nature of the category prediction task. The tuned hyperparameters covered tree depth, learning rate, row and column subsampling rates, minimum child weight, split regularization, and L1/L2 regularization terms. Early stopping on the validation set was applied to

³In the available metadata, the variables *year* and *year_released* contain the same information and therefore do not provide distinct predictive signal. To avoid redundant input features, only *year_released* was retained in the multimodal models.

avoid unnecessary boosting rounds and reduce overfitting. Table 2.2 summarizes the corresponding Optuna search space.

To further boost predictive performance and robustness, we integrated the individual models into a late-fusion ensemble and applied Test-Time Augmentation (TTA). For TTA, each test image underwent 8 independent augmentation passes—incorporating random resized cropping, horizontal flipping, and color jittering—before being processed by the image-based models (ResNet-50, ViT-B/16, and ConvNeXt-Tiny). The softmax probabilities across all 8 passes were then averaged to obtain a more stable and reliable prediction for each model. Finally, these averaged probabilities, along with the predictions from the multimodal XGBoost classifiers, were combined using a weighted average. To determine the optimal ensemble weights, we performed a simplex grid search on the validation set with a step size of 0.05. We systematically evaluated all possible weight combinations that sum to 1.0, and the configuration that maximized the validation Macro-F1 score was selected and subsequently applied to the test set for the final evaluation.

2.5 Results and Model Interpretation

This section presents the empirical results obtained across the different models and pipeline configurations. The primary evaluation metric is Macro-F1, which is particularly appropriate given the imbalanced class distribution. In addition to overall predictive performance, the section also examines confusion matrices, attention-based visualizations, and representative error patterns in order to better understand model behavior.

2.5.1 Overall Performance Comparison

Table 2.3 summarizes the test-set performance across the main modeling strategies considered in this study.

Table 2.3: Overall performance comparison on the **test set** across different models and pipeline configurations. Best results are highlighted in **bold**.

Category	Model / Configuration	Test Accuracy (%)	Test Macro-F1 (%)
Single Modality (Fine-tuned)	ResNet-50 (Baseline)	81.85	78.45
	ViT-B/16	79.26	73.28
	ConvNeXt-Tiny	85.28	81.33
Multi-modal Fusion (embedding + XGBoost)	ResNet-50 Features + Metadata	77.01	69.01
	ViT-B/16 Features + Metadata	76.75	69.11
	DINOv2 Features + Metadata	80.13	73.69
	CLIP Features + Metadata	81.36	75.22
Ensemble Pipeline	Ensemble Base (No TTA)*	88.13	84.50
	Ensemble Base + TTA (8 passes)*	89.14	85.61

*The ensemble combines ResNet-50, ViT-B/16, ConvNeXt-Tiny, and XGBoost models based on CLIP, ResNet-50, ViT-B/16, and DINOv2 features. The displayed results correspond to the final validation-selected weighted ensemble with fusion weights 0.25 (ResNet-50), 0.10 (ViT-B/16), 0.25 (ConvNeXt-Tiny), 0.20 (XGBoost-CLIP), 0.00 (XGBoost-ResNet-50), 0.05 (XGBoost-ViT-B/16), and 0.15 (XGBoost-DINOv2).

Among the single-modality fine-tuned deep learning models, ConvNeXt-Tiny achieved the strongest performance, with a test accuracy of 85.28% and a test macro-F1 of 81.33%. This suggests that the modernized convolutional architecture was particularly suitable for this task, where local visual patterns such as torso prints, accessories, colours, and small texture differences are highly informative for category prediction. ResNet-50 also performed competitively, while ViT-B/16 obtained a slightly lower macro-F1 score, possibly because the dataset size was not large enough for the transformer-based architecture to fully exploit its global-attention advantages after fine-tuning.

The XGBoost-based multi-modal fusion models consistently underperformed relative to the best fine-tuned visual models. This suggests that the main bottleneck is not the lack of utility in metadata such as release year or market value, but rather the representational limitations of fixed visual embeddings. In the present setup, pretrained models such as DINOv2 and CLIP provided frozen features that were robust in a general sense, yet insufficiently adapted to the fine-grained distinctions required to separate visually similar LEGO minifigures. When the underlying visual representation is suboptimal, the addition

of non-leaking metadata cannot fully compensate for the loss of discriminative power.

This point is especially important because the metadata used in this study was deliberately restricted to non-leaking variables. Highly informative categorical variables such as subcategory and theme were excluded because they are strongly related to the target label and would effectively turn the task into a metadata lookup problem. Accordingly, the weaker performance of the fusion models should not be interpreted as evidence that metadata is unhelpful. Rather, metadata appears ineffective here mainly because it is paired with feature extractors that were not adapted to the specific LEGO manifold through task-specific fine-tuning.

The ensemble models achieved the best overall results. The base ensemble without TTA already improved the test accuracy to 88.13% and the macro-F1 score to 84.50%, outperforming the strongest individual model by 2.85 percentage points in accuracy and 3.17 percentage points in macro-F1. This improvement indicates that the different model families captured complementary aspects of the images. For example, CNN-based models are well suited for local visual details, transformer-based models can capture more global relationships, and the XGBoost models based on pretrained embeddings provide an additional non-neural decision mechanism. Averaging their predicted class probabilities therefore reduces the dependence on the failure modes of any single model.

Adding test-time augmentation further improved the ensemble performance to 89.14% accuracy and 85.61% macro-F1. Compared with the base ensemble, TTA increased accuracy by 1.01 percentage points and macro-F1 by 1.11 percentage points. This suggests that the final predictions benefited from averaging over several mildly transformed versions of the same image. Since LEGO minifigure images can vary in alignment, cropping, and small visual details, TTA helps make the prediction less sensitive to a single deterministic test image transformation. The gain is moderate rather than dramatic, which is expected because the models were already trained with data augmentation and the validation/test preprocessing was kept controlled.

Overall, the results show a clear progression from single-model visual classification to model-level aggregation. The best performance was not achieved by adding metadata directly, but by combining diverse visual and embedding-based models through an ensemble. This indicates that, for the present dataset, architectural diversity and prediction averaging were more effective than simple feature-level fusion. The final ensemble with TTA therefore represents the strongest configuration in this study, achieving both the highest test accuracy and the highest macro-F1 score.

2.5.2 Error Analysis

Figure 2.5 presents the normalized confusion matrix of the best-performing model on the test set. Because the matrix is normalized by the true labels, the diagonal entries can be interpreted as class-wise recall values for each category. Overall, the matrix exhibits a clear diagonal structure, indicating that most categories are classified correctly with high recall.

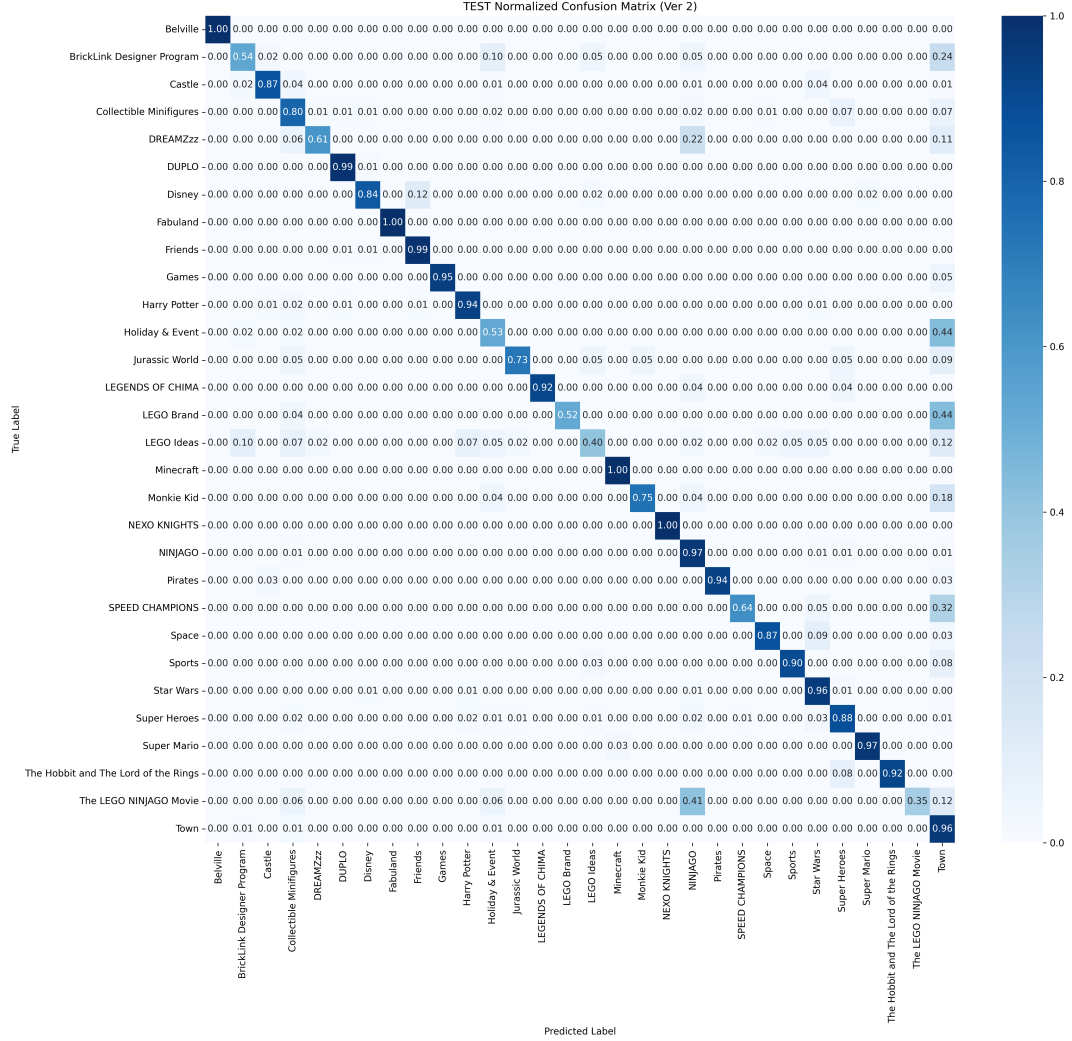


Figure 2.5: Row-normalized confusion matrix on the test set for the best ensemble configuration with TTA ($N = 8$ passes). Darker diagonal cells indicate stronger per-class recall, whereas darker off-diagonal cells reveal systematic confusion between visually similar categories.⁴

As it shown above, several categories achieve near-perfect recall. In particular, Belville,

⁴The confusion matrix was normalized row-wise according to $\text{cm}_{\text{normalized}} = \text{cm} / (\text{row_sums} + 10^{-8})$, where row_sums denotes the sum of the entries in each row of the raw confusion matrix. As a result, each row sums to approximately 1.0 and can be interpreted as the conditional distribution of predicted labels given the true class.

Fabuland, Minecraft, NEXO KNIGHTS, DUPLO, Friends, Super Mario, and Town all obtain diagonal values close to or equal to 1.00. Strong performance is also observed for NINJAGO, Star Wars, Pirates, Harry Potter, The Hobbit and The Lord of the Rings, Sports, Space, Disney, Super Heroes, and Castle, all of which show comparatively high class-wise recall.

At the same time, the confusion matrix also reveals several weaker categories. The lowest recall is observed for The LEGO NINJAGO Movie, whose diagonal entry is approximately 0.35. Other comparatively low-recall categories include LEGO Ideas, LEGO Brand, Holiday & Event, BrickLink Designer Program, DREAMZzz, SPEED CHAMPIONS, and Jurassic World. These classes therefore contribute disproportionately to the remaining gap between strong overall accuracy and imperfect macro-F1, since macro-F1 penalizes low performance on minority or difficult categories more heavily than accuracy alone.

The most pronounced off-diagonal confusion occurs between The LEGO NINJAGO Movie and regular NINJAGO, where a substantial share of The LEGO NINJAGO Movie samples is predicted as NINJAGO. Additional visible confusions include Holiday & Event and LEGO Brand being predicted as Town, Disney being predicted as Friends, and DREAMZzz being predicted as NINJAGO.

However, these off-diagonal concentrations indicate that the remaining classification errors may not be random, but are instead concentrated in a limited set of semantically and visually related categories.

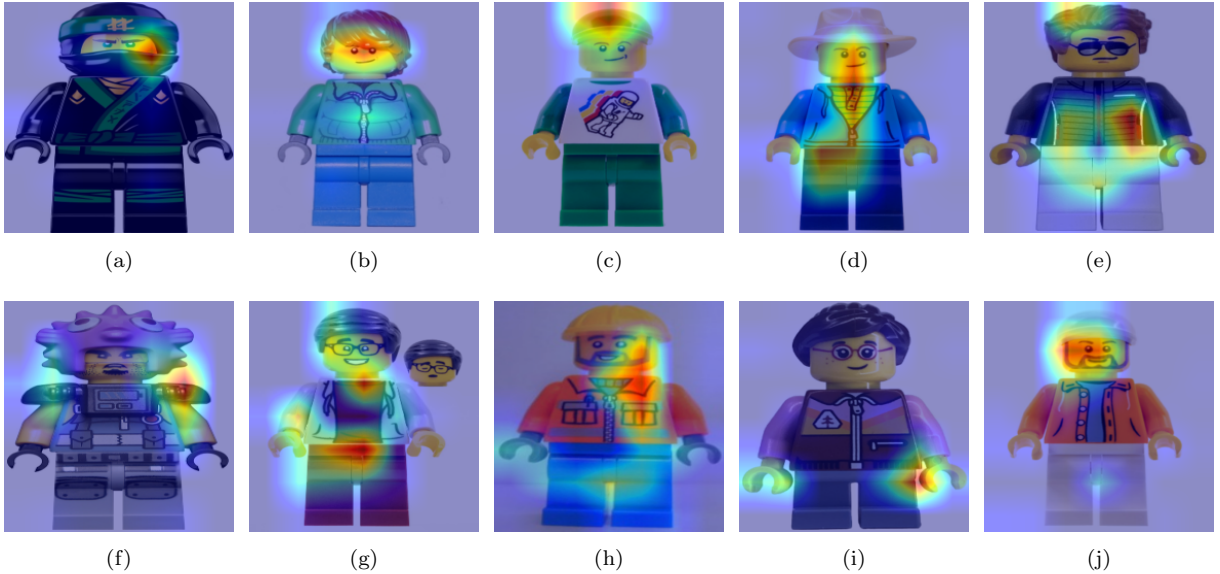


Figure 2.6: Attention heat maps from ConvNeXt-Tiny for ten misclassified test images. Panels are ordered from left to right and top to bottom, and are grouped into five recurring confusion patterns with two representative examples per pattern. Additional attention heat maps are provided in Appendix B.1.⁵

As a first step in the qualitative error analysis, we focus in greater detail on the most frequent and interpretable error patterns rather than attempting to discuss all misclassifications at the same level of depth. Figure 2.6 presents ten representative attention maps, organized into five recurring error patterns with two examples per pattern. These examples are especially informative because they correspond to some of the strongest off-diagonal structures in the confusion matrix and capture different sources of residual classification difficulty, including subtle within-franchise visual similarity, broad thematic overlap, and insufficiently distinctive fine-grained cues. For each selected case, the attention maps in Figure 2.6 highlight which image regions most strongly influenced the prediction. By contrast, the remaining less frequent error types are discussed more briefly at an aggregate level, since they occur only sporadically and are less suitable for detailed case-based interpretation.

The first major error pattern concerns the confusion between *The LEGO NINJAGO Movie* and regular *NINJAGO*, illustrated in panels (a) and (f) of Figure 2.6. The corresponding attention maps are generally convincing: the model focuses on semantically meaningful regions such as the face, mask, headgear, torso, and upper-body costume details. These errors therefore do not appear to be driven by irrelevant background attention or by a failure to localize the minifigure itself. Instead, the difficulty seems to arise from the fine-grained boundary between two highly related categories.

This interpretation is further supported by the structure of the dataset. Among the *The LEGO NINJAGO Movie* examples, 34 out of 90 figures are associated with characters that also belong to the broader *NINJAGO* universe, such as Cole, Dareth, Garmadon, and Jay. This corresponds to approximately 37.8% of the category. Consequently, many images share the same visual language as regular *NINJAGO* figures, including ninja-style clothing, masks, weapons, dark costumes, and closely related character designs. In this setting, predicting *NINJAGO* is visually understandable, even though it remains inconsistent with the more specific dataset label.

In other words, the model appears to capture the broader *NINJAGO* visual identity, but struggles to separate the movie-specific subcategory from the general franchise category. Taken on its own, this error is therefore better understood as a fine-grained label-boundary problem than as a complete classification failure.

The remaining four error patterns share a common structure: figures from *Holiday & Event*, *LEGO Brand*, *BrickLink Designer Program*, and *SPEED CHAMPIONS* are frequently predicted as *Town*. This suggests that *Town* functions as a broad visual sink

⁵Panel details: (a,f) true The LEGO NINJAGO Movie, predicted NINJAGO (0.92, 0.91); (b,g) true Holiday & Event, predicted Town (0.92, 0.91); (c,h) true LEGO Brand, predicted Town (0.79, 0.25); (d,i) true BrickLink Designer Program, predicted Town (0.70, 0.19); and (e,j) true SPEED CHAMPIONS, predicted Town (0.89, 0.80).

category for generic civilian-like minifigures. Many examples from these categories do not contain strongly distinctive clothing, accessories, or theme-specific visual markers. Their true labels may depend on event context, brand context, production origin, or the surrounding set rather than on the isolated minifigure image itself. Consequently, the model’s prediction of *Town* is often visually plausible, even when it is inconsistent with the dataset annotation.

The attention maps support this interpretation in most cases. Several misclassified examples still show activations on semantically meaningful regions such as the face, hair, torso, arms, and clothing, indicating that the model is not mainly relying on irrelevant background artifacts. However, a small number of cases show weaker attention patterns. For example, in panels (i) and (c), the strongest activation is concentrated around less discriminative regions such as the hand or hat. Although these regions are still part of the minifigure, they are less informative than facial details, torso prints, or full costume patterns for category recognition. These examples suggest that the model may occasionally rely on salient but weak local cues.

Therefore, these errors reflect two interacting mechanisms. The dominant mechanism is semantic ambiguity: the true labels are not always visually self-contained, while *Town* provides a visually plausible alternative for generic figures. A secondary mechanism is imperfect attention in a few hard cases, where limited or ambiguous visual evidence may lead the model to focus on less discriminative local regions. Importantly, these weak attention examples should be interpreted as local failure cases rather than evidence of systematic model failure.

Across both major error types, limited sample size likely amplifies the observed instability. Several difficult categories contain only a small number of examples, so the 70/15/15 split leaves relatively few training images from which to learn distinctive category-specific cues and only a small number of validation and test images on which performance is assessed. In such settings, class-wise recall becomes highly sensitive to individual mistakes, and small differences in visual composition can have a disproportionate effect on the reported metrics. Thus, the weaker performance of categories such as *The LEGO NINJAGO Movie*, *Holiday & Event*, *LEGO Brand*, *BrickLink Designer Program*, and *SPEED CHAMPIONS* should be interpreted not only in terms of semantic overlap and fine-grained ambiguity, but also in light of limited training variation and unstable small-sample evaluation.

Several additional low-recall categories further illustrate that weak class-wise performance does not have a single explanation. In particular, *LEGO Ideas* and *BrickLink Designer Program* are difficult because they are not visually coherent themes in the same sense as *Minecraft*, *NINJAGO*, or *DUPLO*. Instead, they are closer to product-line, submission-origin, or production-context categories. *LEGO Ideas* contains fan-submitted sets from

many different genres and visual worlds, while *BrickLink Designer Program* similarly consists of fan-designed projects selected through a separate design and production process.⁶ As a result, minifigures from these categories do not necessarily share a consistent visual style. Some may resemble ordinary civilian figures, while others may look closer to historical, fantasy, space, promotional, or town-like characters. Therefore, when the model confuses these categories with visually plausible alternatives such as *Town*, the error often reflects a mismatch between a context-based label and the visual information available in the isolated image. In other words, the model is being asked to recover a category that may be defined more by product origin than by appearance.

The case of *DREAMZzz* is somewhat different. Unlike *LEGO Ideas* or *BrickLink Designer Program*, *DREAMZzz* is closer to a visually meaningful theme and does contain distinctive fantasy-oriented elements, such as unusual headgear, colorful designs, dream-like creatures, and action-oriented costumes. However, its weaker performance is likely driven by limited sample size and high intra-class variation. With relatively few examples, the model may not observe enough variation during training to learn stable *DREAMZzz*-specific features. In addition, some of its visual cues may overlap with other fantasy or action-oriented categories, such as *NINJAGO*, *NEXO KNIGHTS*, or similar highly stylized themes. Thus, the difficulty of *DREAMZzz* is better interpreted as a small-sample and feature-overlap problem rather than as a purely context-defined label problem.

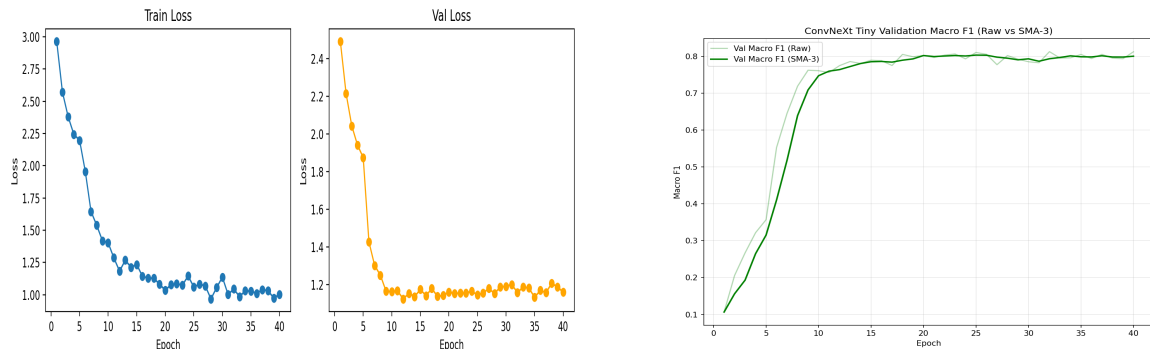
This distinction is important because low recall should not be interpreted uniformly across categories. For *LEGO Ideas* and *BrickLink Designer Program*, the core issue is that the label is only weakly visually observable, since the category reflects submission or production context rather than a unified design language. For *DREAMZzz*, the label is more visually meaningful, but the available training data may be insufficient for the model to learn a stable category boundary. Therefore, the remaining errors reflect a combination of context-dependent labels, limited sample size, and overlap between visually related themes, rather than a simple failure of the model architecture.

2.6 Discussion and Conclusion

The results of this assignment show that the proposed pipeline achieved strong overall performance on the LEGO Minifig category classification task. Starting from exploratory data analysis, the study first identified two central challenges in the dataset: class imbalance and substantial semantic overlap between categories, subcategories, and themes. Based on these observations, the task was restricted to the 30 most frequent categories

⁶See the official LEGO Ideas platform: <https://ideas.lego.com/>. For BrickLink Designer Program, see: <https://www.bricklink.com/v3/designer-program/main.page>.

and evaluated primarily using macro-F1, which gives equal importance to each class. Several modelling strategies were then compared, including fine-tuned deep learning models, metadata-augmented XGBoost classifiers based on pretrained image embeddings, and a final late-fusion ensemble with test-time augmentation. Among the individual models, ConvNeXt-Tiny achieved the strongest single-model performance, while the final ensemble with TTA obtained the best overall result. This suggests that architectural diversity and probability-level model aggregation were more effective than relying on any single model or simple metadata fusion alone.



(a) ConvNeXt-Tiny training and validation loss curves.

(b) Smoothed validation macro-F1 curve for ConvNeXt-Tiny.

Figure 2.7: Training dynamics of the strongest single-model architecture, ConvNeXt-Tiny. The left panel shows the training and validation loss curves, while the right panel shows the smoothed validation macro-F1 trajectory used to assess stability during training.

Although some degree of overfitting was observed during training, the final results suggest that the models retained strong generalization ability. This can be assessed through the training and validation loss curves, the macro-F1 trajectories, and the final train-validation-test performance statistics. As shown in Figure 2.7, the training and validation losses diverge to some extent, yet the validation macro-F1 remains comparatively stable rather than collapsing. In particular, a gap between training and validation performance indicates that the models were able to fit the training data more closely than the unseen validation data, which is expected in a high-capacity deep learning setting with visually heterogeneous classes. However, the validation macro-F1 did not collapse, and the final test-set performance remained strong, especially for the ensemble model. This indicates that the regularization strategies used in the pipeline, including data augmentation, random erasing, CutMix, dropout, label smoothing, AdamW weight decay, early stopping, and test-time augmentation, helped control overfitting. Therefore, the observed overfitting should be interpreted as a manageable tendency rather than a failure of generalization.

The remaining limitations are closely related to the structure and quality of the dataset itself. First, several categories contain relatively few observations, which makes both

learning and evaluation unstable. For the smallest retained categories, the validation and test sets contain only a limited number of examples, meaning that a small number of misclassifications can substantially affect class-wise recall and macro-F1. Second, some categories are visually and semantically overlapping, such as The LEGO NINJAGO Movie and regular NINJAGO, where many figures share similar characters, clothing styles, masks, weapons, and franchise-specific design patterns. Third, some labels are not fully visually self-contained. Categories such as LEGO Ideas, BrickLink Designer Program, LEGO Brand, and Holiday & Event may depend more on product origin, event context, or set-level information than on the isolated appearance of the minifigure. As a result, part of the remaining error is not simply caused by model weakness, but by the fact that the image alone may not contain enough information to recover the dataset label reliably.

Although the current pipeline already uses extensive data augmentation and regularization, the remaining errors suggest that generic image-level augmentation alone is insufficient to resolve the main limitations of the task. Future work could therefore focus on more targeted strategies. First, class-aware training objectives could be introduced to make the optimization process more consistent with the macro-F1 evaluation criterion. For example, weighted cross-entropy, class-balanced loss, or focal loss could assign greater penalty to errors on underrepresented or consistently low-recall categories, encouraging the model to improve performance on difficult classes rather than mainly optimizing the majority classes.

Second, hard-negative training could be used to address the most persistent confusion patterns observed in the confusion matrix. Category pairs that are repeatedly confused, such as The LEGO NINJAGO Movie versus NINJAGO or Town versus more generic civilian-like categories, could be oversampled or used in contrastive learning objectives. This would force the model to learn finer distinctions between visually similar but label-distinct categories.

Third, a hierarchical classification strategy could be explored. Instead of directly predicting one of all target categories in a single step, the model could first classify images into broader visual families and then make a more fine-grained prediction within each family. This would be particularly useful because many errors occur within semantically related groups rather than randomly across all categories.

Finally, future work could introduce region-guided or part-based modeling. Rather than processing the full image uniformly, the minifigure could first be detected or segmented, and important regions such as the head, torso, legs, and accessories could be encoded separately. This would restrict the model’s effective attention space to semantically meaningful regions and reduce reliance on weak local cues or background artifacts. Taken together, these extensions would address the remaining errors more directly than simply adding

more generic data augmentation.

Chapter 3

Assignment 3

3.1 Intro

3.2 Implementation

3.3 Conclusion

Chapter 4

Assignment 4: Graph Analytics on the Epstein Communication Network

4.1 Objective and Analytical Strategy

The objective of this analysis is to investigate the structural organization of the Epstein-related communication network extracted from the graph database provided for the assignment. Rather than immediately applying advanced graph-mining algorithms, the analysis follows a progressive exploratory workflow. First, the communication graph is extracted and cleaned from the larger Memgraph database. Second, exploratory graph analysis is used to identify large-scale structural patterns. Third, increasingly advanced graph-theoretical measures are introduced to quantify the observations obtained during the exploratory phase. Finally, the structural properties of the network are interpreted through visualization, robustness analysis, and community detection techniques.

The central research question is therefore formulated as follows:

Does the extracted communication network behave like a decentralized social network with several communities, or is it structurally dominated by a small number of highly central actors?

To answer this question, the analysis combines exploratory visualization, graph diagnostics, centrality analysis, diffusion-based ranking, structural cohesion analysis, robustness simulations, and modularity-based community detection.

4.2 Graph Extraction and Cleaning

The original Memgraph database contains several node and edge types, including documents, claims, entities, topics, tags, and locations. Since the purpose of this analysis is to study communication and social proximity, the graph extraction focuses exclusively on **Entity** nodes connected through **RELATED_TO** relationships.

Only three interaction types were retained:

- sent email to,
- sent message to,
- met with.

These relationships were selected because they correspond to direct communication or observable social interaction. More ambiguous or weakly interpretable relationships were excluded in order to obtain a focused and interpretable communication graph.

The following Cypher query was used to extract the graph:

```
MATCH (a:Entity)-[r:RELATED_TO]->(b:Entity)
WHERE r.action IN [
    "sent email to",
    "sent message to",
    "met with"
]
AND NOT a.name CONTAINS "unknown"
AND NOT b.name CONTAINS "unknown"
RETURN a.name AS source,
       b.name AS target,
       r.action AS action
LIMIT 5000;
```

After filtering, the graph contained 724 nodes and 852 directed edges. Additional preprocessing removed raw email-address entities and duplicate interaction artifacts. The final cleaned graph used for structural analysis contained 718 nodes and 720 edges.

A directed edge from node i to node j indicates that an extracted interaction exists from entity i to entity j .

Formally, the graph is represented as

$$G = (V, E),$$

where V ($|V| = 724$) is the set of entities and E ($|E| = 852$) is the set of directed interaction edge

4.3 Exploratory Network Analysis

The first analytical step consisted of visualizing the extracted graph in Memgraph Lab in order to identify large-scale structural patterns before applying formal graph metrics.

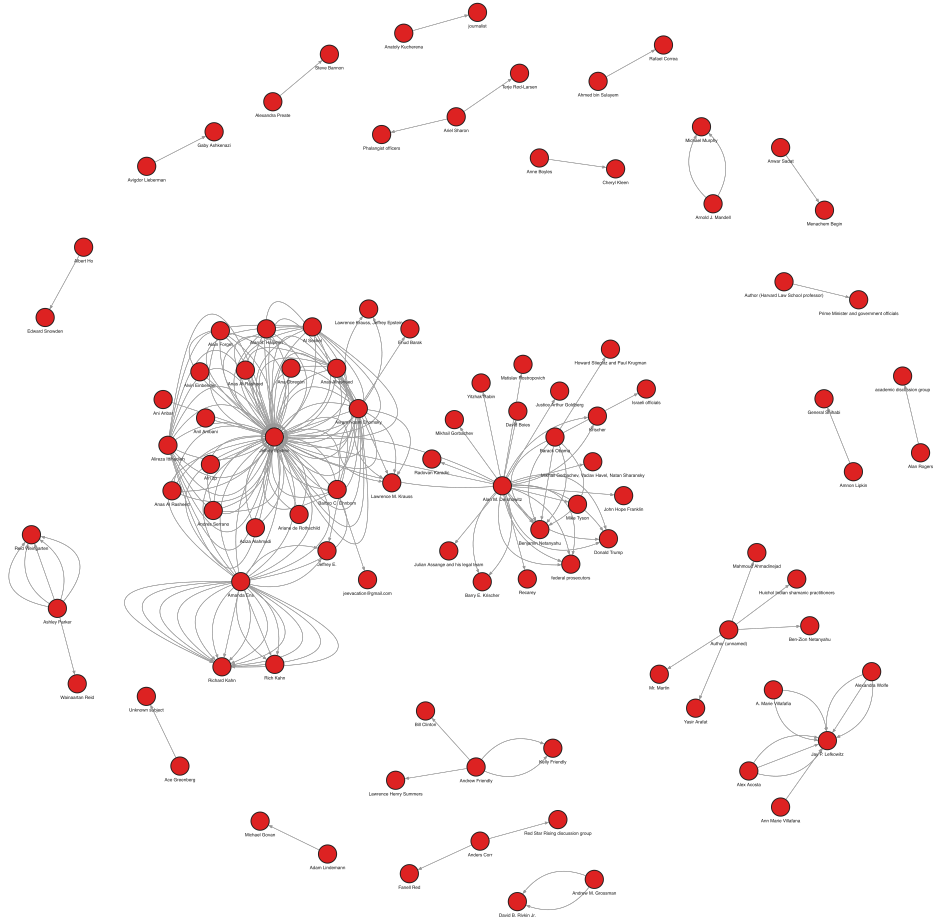


Figure 4.1: Exploratory visualization of the extracted communication graph in Memgraph Lab.

The exploratory visualization immediately suggested that the graph was not uniformly connected. Instead, the network appeared highly sparse, with a dominant central region surrounded by many peripheral nodes and disconnected fragments.

Several important observations emerged from this visualization. First, the graph appeared strongly centralized around a limited number of nodes. Second, many nodes were located at the periphery of the graph and appeared weakly connected to the dominant structure. Third, the visualization did not reveal the presence of clearly separated large communities.

These exploratory observations motivated the following hypothesis:

The extracted communication network is structurally dominated by a central hub, while a limited number of secondary nodes act as bridges between peripheral regions of the graph.

This hypothesis motivated the use of graph diagnostics, Personalized PageRank, betweenness centrality, k-core decomposition, robustness analysis, and modularity-based community detection.

4.4 Global Network Diagnostics

After the exploratory visualization phase, additional graph-level diagnostics were computed in Python using the `NetworkX` library in order to quantitatively characterize the structural properties that had initially been observed visually.

More specifically, the following global graph diagnostics were computed:

- the number of nodes and edges;
- graph density;
- average node degree;
- connected-component decomposition;
- giant-component size;
- degree distribution;
- k-core decomposition;
- robustness under node removal.

These diagnostics provide complementary information about the organization of the network at different scales. While exploratory visualization allows qualitative inspection of connectivity patterns, graph diagnostics quantify sparsity, fragmentation, structural cohesion, and dependence on highly connected actors.

Since the analysis was performed after graph cleaning and preprocessing (removal of duplicate relationships, missing labels, and selected entity filtering), we denote the cleaned graph by

$$\tilde{G} = (\tilde{V}, \tilde{E}),$$

where:

- $\tilde{V}$ denotes the set of cleaned nodes,
- $\tilde{E}$ denotes the set of cleaned edges.

The final cleaned graph contained

$$|\tilde{V}| = 718$$

nodes and

$$|\tilde{E}| = 720$$

edges.

A first global measure of interest is the graph density, defined for an undirected graph as

$$\delta(\tilde{G}) = \frac{2|\tilde{E}|}{|\tilde{V}|(|\tilde{V}| - 1)}.$$

Substituting the observed values yields

$$\delta(\tilde{G}) = \frac{2 \times 720}{718 \times 717} \approx 0.0028.$$

Graph density measures the proportion of existing edges relative to the total number of edges that would exist in a fully connected graph.

More formally,

$$\delta(\tilde{G}) \in [0, 1],$$

where:

- $\delta(\tilde{G}) = 0$ corresponds to an empty graph,
- $\delta(\tilde{G}) = 1$ corresponds to a complete graph.

The observed value of 0.0028 indicates that only approximately 0.28% of all theoretically possible interactions are present.

This result is important because it immediately characterizes the communication network as highly sparse.

From a structural perspective, sparsity implies that interactions are concentrated among a relatively small subset of entities rather than being uniformly distributed throughout the network. Sparse communication networks are often associated with strong heterogeneity, the emergence of hubs, and fragmented peripheral structures.

A second important quantity is the average node degree, defined as

$$\langle k \rangle = \frac{1}{|\tilde{V}|} \sum_{v \in \tilde{V}} k_v.$$

Substituting the observed values gives

$$\langle k \rangle \approx 2.01.$$

The average degree represents the expected number of neighbors of a randomly selected node.

In practical terms, this means that a typical node in the graph interacts directly with approximately two other entities.

However, this average must be interpreted together with the maximum observed degree:

$$k_{\max} = 278.$$

The difference between $\langle k \rangle$ and $k_{\max}$ is substantial.

This disparity reveals that the graph is not homogeneous. Most entities participate in only

a very limited number of interactions, whereas a very small number of nodes concentrate a disproportionately large fraction of the connectivity.

This type of structural imbalance is characteristic of hub-dominated networks and provides the first quantitative indication that the communication graph may exhibit strong centralization effects.

This observation directly motivates the subsequent degree-distribution and centrality analyses.

Metric	Value
Number of nodes	718
Number of edges	720
Density	0.0028
Average degree	2.01
Maximum degree	278
Number of connected components	60
Largest component size	574
Largest component share	79.9%
Maximum core number	3

Table 4.1: Global structural diagnostics of the cleaned communication graph.

The graph contains 60 connected components. However, the largest connected component contains 574 nodes, corresponding to approximately 79.9% of the graph.

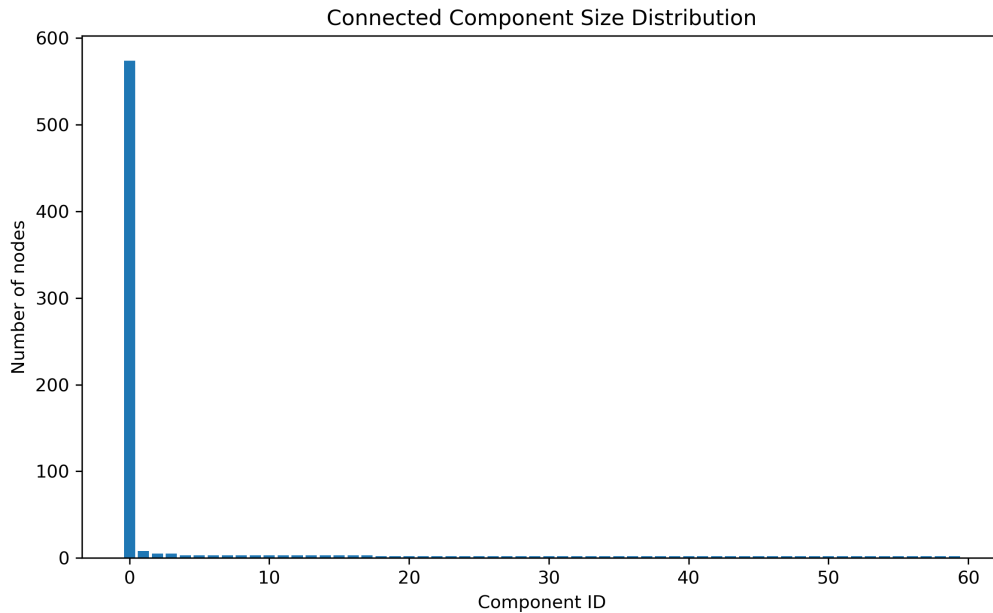


Figure 4.2: Connected component size distribution of the cleaned communication graph.

Figure 4.2 shows that the graph is not fragmented into many similarly sized groups.

Instead, it is composed of one dominant giant component together with many small peripheral fragments.

This giant-component structure is particularly important for diffusion-based methods such as Personalized PageRank because random walks can only propagate within connected regions of the graph. Nodes outside the dominant component are therefore structurally disconnected from the main communication structure.

4.5 Degree Distribution and Hub Structure

The previous global diagnostics established that the cleaned communication graph is sparse and structurally heterogeneous.

Although sparsity alone does not imply centralization, the large difference observed between the average degree $\langle k \rangle$ and the maximum observed degree $k_{\max}$ already suggests that interactions may be concentrated around a limited number of actors. To investigate this phenomenon more directly, the degree distribution was analyzed. The degree of a node measures the number of direct interactions associated with that node.

For the cleaned graph $\tilde{G} = (\tilde{V}, \tilde{E})$, the degree of a node $v \in \tilde{V}$ is defined as the number of adjacent nodes.

Using the adjacency matrix representation, the degree can be written as

$$k_v = \sum_{u \in \tilde{V}} \tilde{A}_{vu},$$

where the adjacency matrix $\tilde{\mathbf{A}} = (\tilde{A}_{ij})$ is defined element-wise as $\tilde{A}_{ij} = \begin{cases} 1, & (v_i, v_j) \in \tilde{E}, \\ 0, & \text{otherwise.} \end{cases}$

Consequently, the degree corresponds to the sum of the entries of the row associated with node v .

Degree analysis provides a local notion of importance because it only considers immediate neighbors. Therefore, if a node exhibits large degree, this indicates that it directly participates in many interactions.

Figure 4.3 shows the empirical degree distribution.

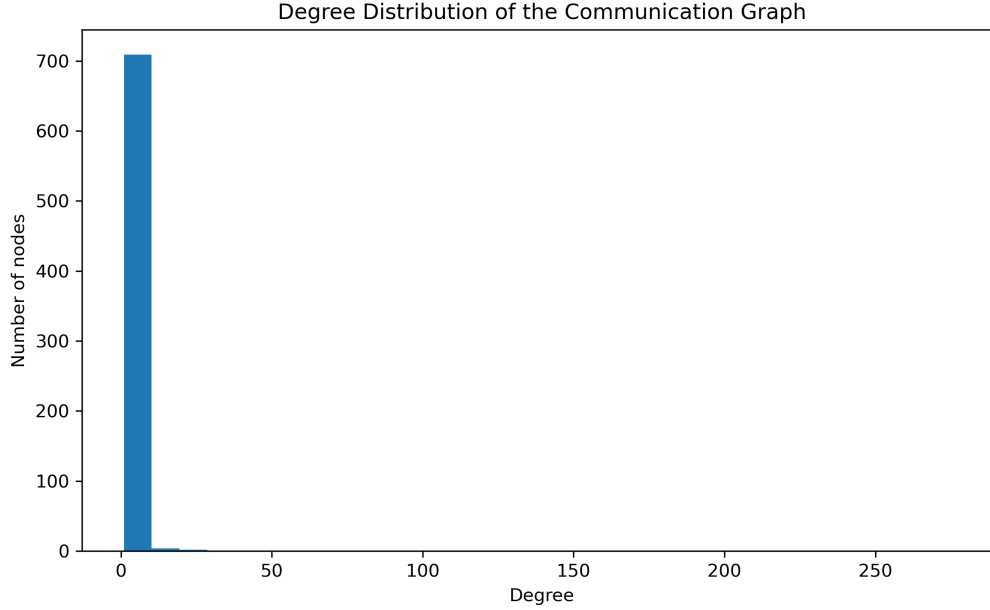


Figure 4.3: Degree distribution of the cleaned communication graph.

The distribution is strongly right-skewed. Most nodes have degree close to one, whereas a very small number of nodes exhibit substantially larger degree values. This pattern is characteristic of hub-dominated interaction networks.

The most striking example is Jeffrey Epstein, whose degree equals 278, which is approximately 138 times larger than the average degree. This result provides quantitative support for the exploratory observations and suggests that direct interactions are concentrated around a small subset of structurally dominant nodes. However, degree remains a purely local quantity. Two nodes may have identical degree while occupying completely different structural positions in the network.

Degree does not account for:

- indirect paths,
- information propagation,
- structural reachability,
- intermediary roles.

Consequently, the next analytical step moves beyond local connectivity and investigates centrality measures capable of capturing global structural influence.

4.6 Centrality Analysis

The previous analyses revealed that the communication graph exhibits three important structural characteristics. First, the graph is sparse. Second, interactions are strongly concentrated around a small number of hubs. Third, the network contains one dominant connected component. Together, these observations suggest that understanding structural importance requires more than counting direct interactions. The next objective is therefore to study how influence propagates through the network. Two complementary perspectives are considered. The first perspective studies diffusion and structural proximity. The second perspective studies brokerage and intermediary roles. These objectives are addressed respectively through Personalized PageRank and betweenness centrality.

4.7 Personalized PageRank

The strong hub structure observed during the exploratory analysis suggests that influence may not be determined solely by direct interactions. In particular, highly connected nodes may affect not only their immediate neighbors but also more distant regions of the network through indirect paths. To capture this phenomenon, Personalized PageRank (PPR) was computed. Personalized PageRank extends the classical PageRank algorithm by introducing a preferred restart distribution centered around a selected seed node. Rather than measuring global popularity, Personalized PageRank estimates how structurally reachable each node is under a random walk repeatedly restarted from the selected actor. This approach is particularly relevant in the present setting because the objective is not to rank all actors globally but to characterize the structural neighborhood surrounding Jeffrey Epstein.

Let $\mathbf{r}$ denote the Personalized PageRank vector.

The stationary distribution satisfies:

$$\mathbf{r} = \alpha \mathbf{M}^T \mathbf{r} + (1 - \alpha) \mathbf{e}_s,$$

where $\mathbf{M}$ is the transition matrix of the cleaned graph, α is the damping factor, $\mathbf{e}_s$ is the personalization vector and s denotes the selected seed node.

The term

$$\alpha \mathbf{M}^T \mathbf{r}$$

models the diffusion process across edges, while

$$(1 - \alpha)\mathbf{e}_s$$

forces periodic returns toward the selected actor.

In this analysis, Jeffrey Epstein was selected as the personalization seed and the damping factor was fixed to $\alpha = 0.85$. This value is commonly used because it balances local exploration and long-range diffusion.

The Personalized PageRank results reveal an extremely concentrated stationary distribution. More specifically, the score associated with Jeffrey Epstein was $r_{\text{Jeffrey Epstein}} = 0.5103$, whereas the second-largest value was obtained by $r_{\text{jeevacation@gmail.com}} = 0.0046$. The difference between these values spans more than two orders of magnitude. This result indicates that the diffusion process remains overwhelmingly concentrated around the selected seed node. In other words, random-walk trajectories repeatedly return to the same structural region of the graph rather than spreading uniformly across the network. Importantly, Personalized PageRank measures structural reachability within the communication graph. It should therefore be interpreted as a network property and not as a measure of responsibility, influence, or real-world importance.

4.8 Betweenness Centrality and Brokerage Structure

Although Personalized PageRank captures diffusion processes, it does not explicitly quantify whether nodes act as intermediaries connecting different regions of the network.

To study this phenomenon, betweenness centrality was computed.

Since the previous analysis showed that Personalized PageRank was heavily concentrated, only the top 100 nodes according to PPR were retained.

This filtering step serves two purposes.

First, it reduces peripheral noise.

Second, it restricts the analysis to the structurally relevant region identified during diffusion analysis.

Betweenness centrality evaluates how frequently a node appears on shortest paths between other nodes.

For a node v , the measure is defined as:

$$BC(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}},$$

where: σ_{st} denotes the total number of shortest paths between nodes s and t , $\sigma_{st}(v)$ denotes the number of those paths passing through node v . Shortest paths were computed using the shortest-path procedure implemented internally by the Brandes algorithm, which computes exact betweenness centrality efficiently for sparse graphs.

Conceptually, shortest paths represent the minimum number of interactions required to connect two entities.

Nodes with high betweenness therefore occupy structurally strategic positions because they connect otherwise distant parts of the graph.

The resulting scores are summarized below.

Node	Personalized PageRank	Betweenness
Jeffrey Epstein	0.5103	0.9910
jeevacation@gmail.com	0.0046	0.0000
Alan M. Dershowitz	0.0037	0.0001
Darren K. Indyke	0.0034	0.0006
Richard Kahn	0.0031	0.0001
Bill Clinton	0.0028	0.0001
Ehud Barak	0.0027	0.0003
Martin G. Weinberg	0.0026	0.0002

Table 4.2: Comparison between Personalized PageRank and betweenness centrality.

The computed betweenness centrality for Jeffrey Epstein was

$$BC(\text{Jeffrey Epstein}) = 0.9910.$$

This value is extremely close to the theoretical maximum value of one. Therefore, almost all shortest paths inside the focused subgraph pass through the same dominant structural intermediary. This result indicates that Jeffrey Epstein does not only occupy a highly connected position (high degree), but also acts as the principal bridge through which communication paths are organized.

This result reinforces the conclusions of the previous analyses. The graph is not only diffusion-centered around a dominant actor, but shortest-path connectivity is also concentrated around the same hub. Secondary nodes exhibit non-zero betweenness values but remain substantially smaller, suggesting local bridge roles rather than global brokerage.

4.9 Final Network Visualization

The top-Personalized-PageRank subgraph was exported to Gephi for visualization. Node size was mapped to Personalized PageRank, meaning that larger nodes are more structurally reachable from Jeffrey Epstein. Node color was mapped to betweenness centrality, allowing brokerage nodes to be visually distinguished.

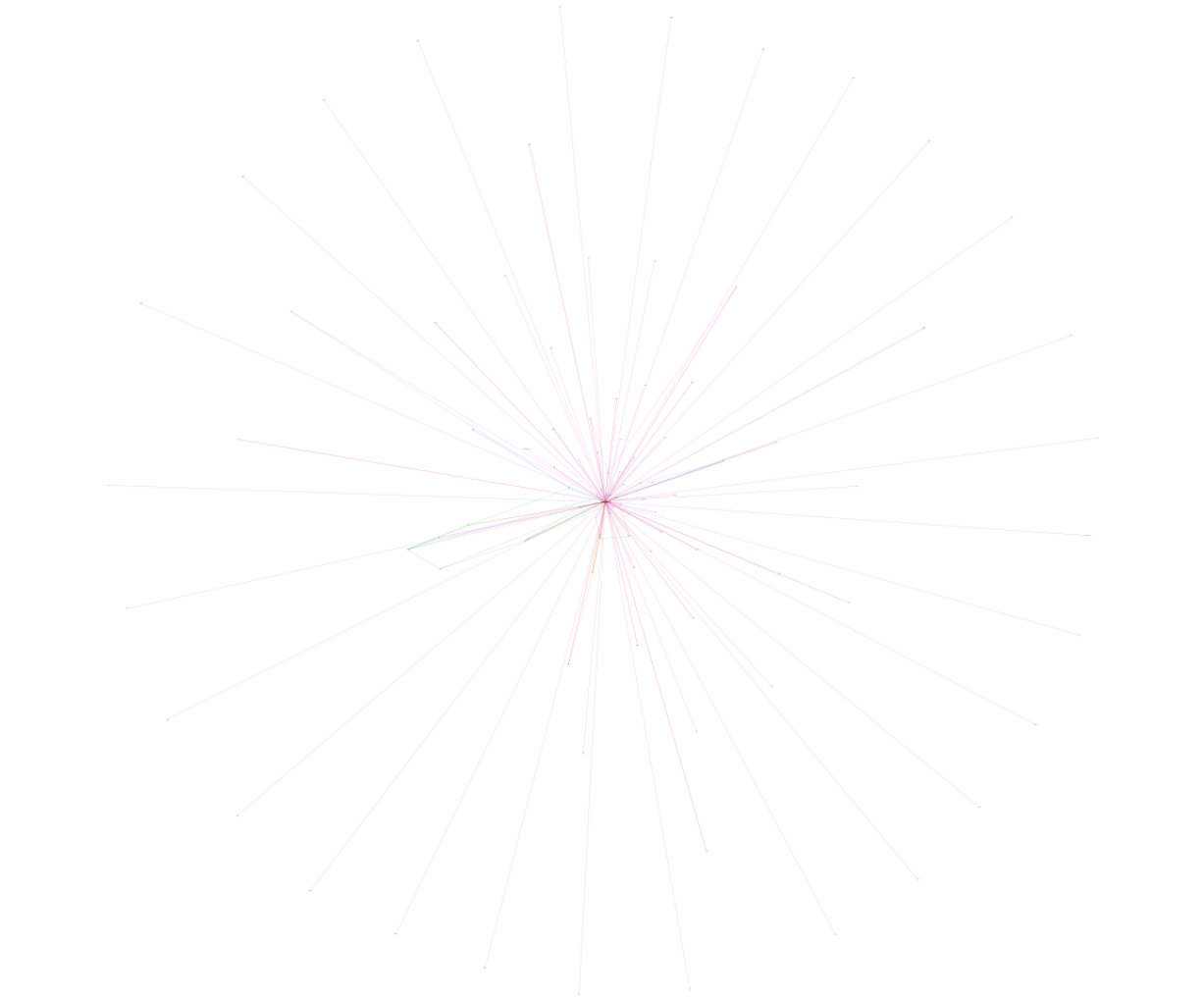


Figure 4.4: Global visualization of the top-Personalized-PageRank communication subgraph. Node size represents Personalized PageRank and node color represents betweenness centrality.

Figure 4.4 confirms the strong centralization of the communication network. The graph displays a hub-and-spoke structure in which many peripheral nodes are connected through one dominant central region.

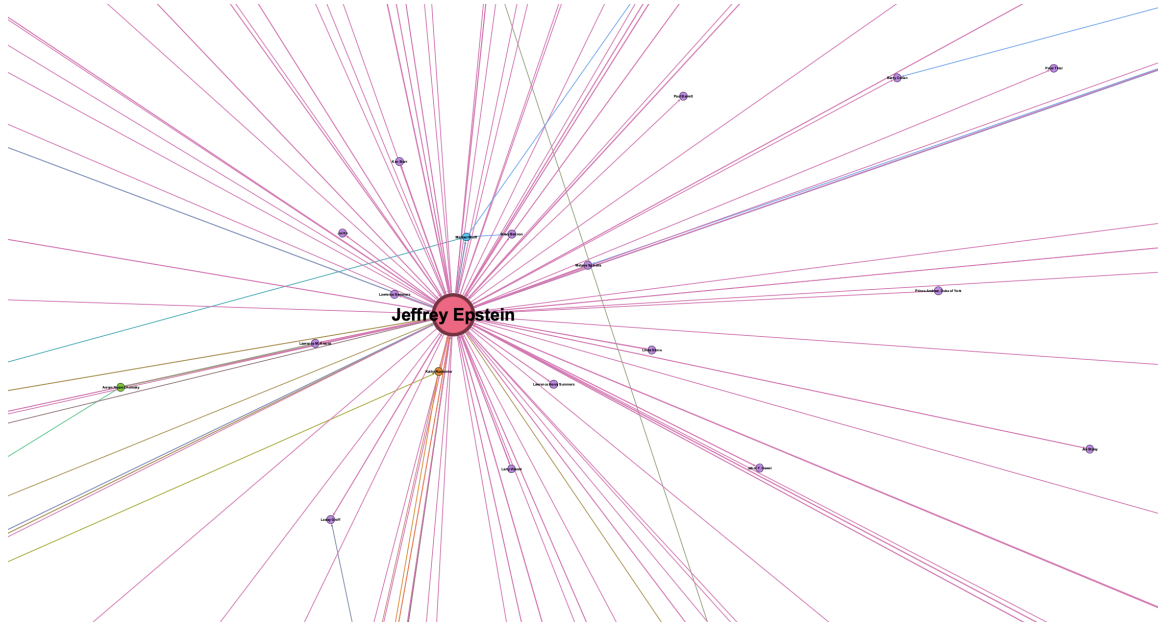


Figure 4.5: Zoomed visualization of the central region of the communication graph.

Figure 4.5 provides a closer view of the central region. The dominant structural position of Jeffrey Epstein becomes immediately visible, while several secondary bridge nodes also emerge.

The visualization further supports the interpretation that the network behaves as a highly centralized interaction structure rather than a balanced multi-community social network.

4.10 Peripheral Components and Core–Periphery Structure

Besides the dominant hub-and-spoke structure around the central node, the visualization also reveals several disconnected peripheral components. These components consist of isolated dyads, triads, or very small interaction groups disconnected from the main Epstein-centered component.

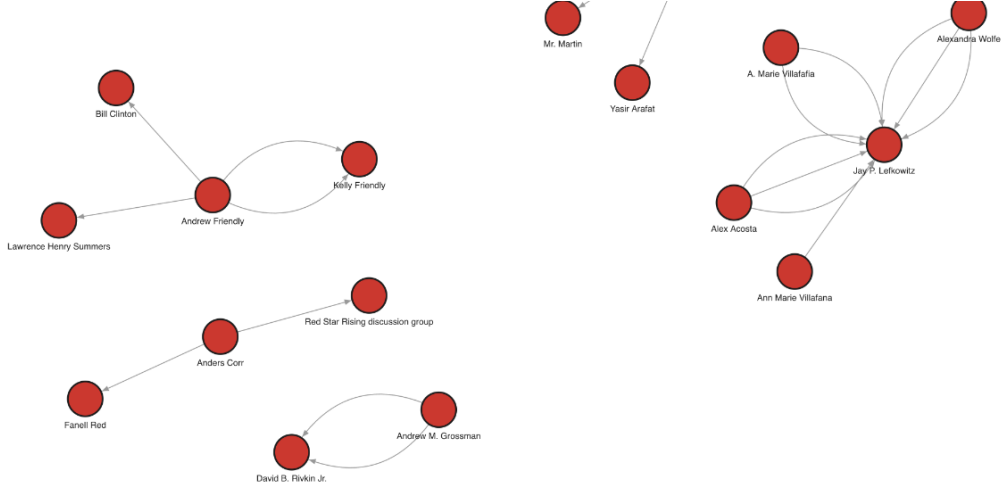


Figure 4.6: Peripheral disconnected components in the communication graph.

From a graph-theoretical perspective, the communication graph is therefore not composed of one fully connected network. Instead, it can be represented as a union of connected components:

$$G = \bigcup_{k=1}^K C_k,$$

where each C_k denotes one connected component.

The dominant component contains the main hub structure, while the remaining components form structurally peripheral fragments. This observation is important because diffusion-based methods such as Personalized PageRank cannot naturally propagate between disconnected components. If no path exists between two components, then a random walk starting in one component can never reach nodes in the other component.

The existence of these disconnected peripheral structures therefore further supports the interpretation that the graph behaves as a sparse core-periphery network rather than a decentralized social structure.

4.11 k-core Decomposition

To further investigate the internal structure of the graph, a k-core decomposition was computed. A k-core is a maximal subgraph in which every node has at least degree k within that subgraph.

Unlike degree centrality, which only measures direct connectivity, k-core decomposition

identifies nodes that are embedded inside structurally cohesive regions of the network.

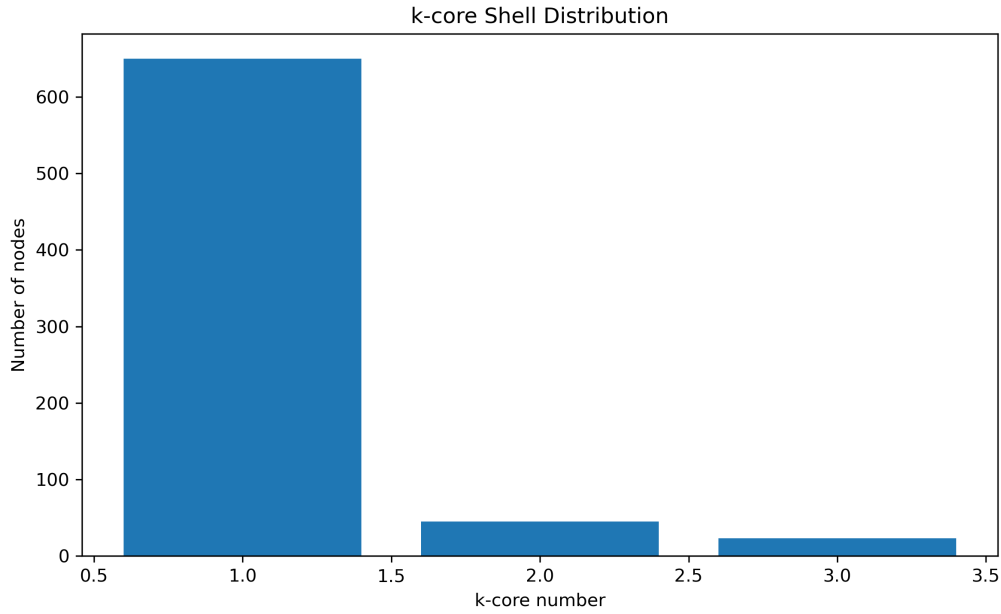


Figure 4.7: k-core shell distribution of the cleaned communication graph.

Figure 4.7 shows that most nodes belong to the outer 1-core shell. Only a limited number of nodes survive inside the deeper 2-core and 3-core regions.

The maximum observed core number is equal to 3, indicating that the graph contains only a shallow structurally cohesive nucleus.

Node	k-core number
Darren K. Indyke	3
Hillary Rodham Clinton	3
Andrew Friendly	3
Bill Clinton	3
Lawrence Henry Summers	3
Jeffrey E.	3
Richard Kahn	3
Kelly Friendly	3
Amanda Ens	3
Michael Wolff	3
Donald Trump	3
Avram Noam Chomsky	3
Lawrence M. Krauss	3
Benjamin Netanyahu	3
Martin G. Weinberg	3

Table 4.3: Examples of nodes located in the deepest observed k-core.

The k-core results refine the interpretation of the communication network. Although

the graph contains a small structurally cohesive nucleus, this nucleus remains shallow and highly limited in size. The network is therefore better described as a sparse core–periphery structure than as a dense multi-community system.

4.12 Network Robustness and Hub Dependence

The previous analyses strongly suggest that the network depends heavily on a small number of dominant hubs. To test this hypothesis more directly, a robustness simulation was conducted.

Two node-removal strategies were compared. In the first strategy, nodes were removed randomly. In the second strategy, nodes were removed in descending order of degree, meaning that the most connected hubs were removed first. After each removal step, the size of the largest connected component was measured.

This experiment evaluates whether the graph is robust to random failures but fragile to targeted hub removal.

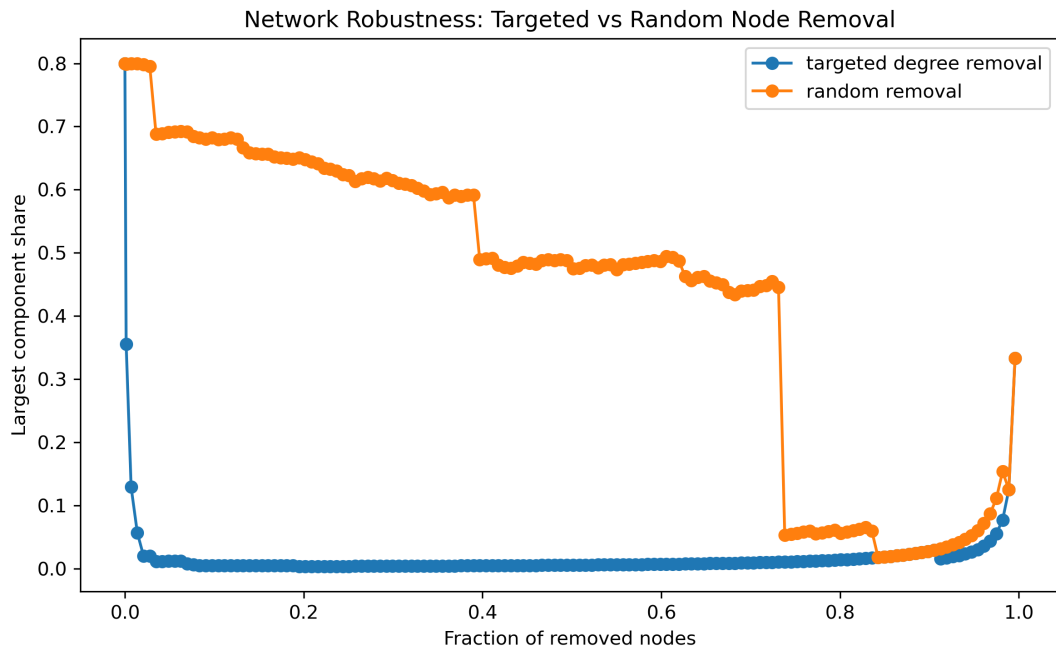


Figure 4.8: Network robustness simulation comparing random node removal with targeted removal of high-degree nodes.

Figure 4.8 reveals a clear contrast between the two removal strategies. Under random node removal, the size of the largest connected component decreases gradually. This indicates that random failures affecting peripheral nodes do not immediately destroy the global structure of the graph.

In contrast, targeted removal of high-degree hubs causes the largest connected component to collapse almost immediately. The communication network is therefore highly vulnerable to the removal of structurally central actors.

This robustness experiment significantly strengthens the central conclusion of the analysis. The graph is not merely visually centralized; it is functionally dependent on its dominant hubs. Once these hubs are removed, the global communication structure rapidly fragments.

4.13 Modularity and Community Structure

To determine whether the graph contains meaningful community structure, modularity analysis was performed in Gephi using the Louvain method.

Modularity measures whether a graph can be partitioned into groups with dense internal connectivity and relatively sparse external connectivity.

The resulting modularity score was

$$Q = 0.082.$$

Gephi detected three communities. However, the modularity value remains extremely low, indicating weak community structure.

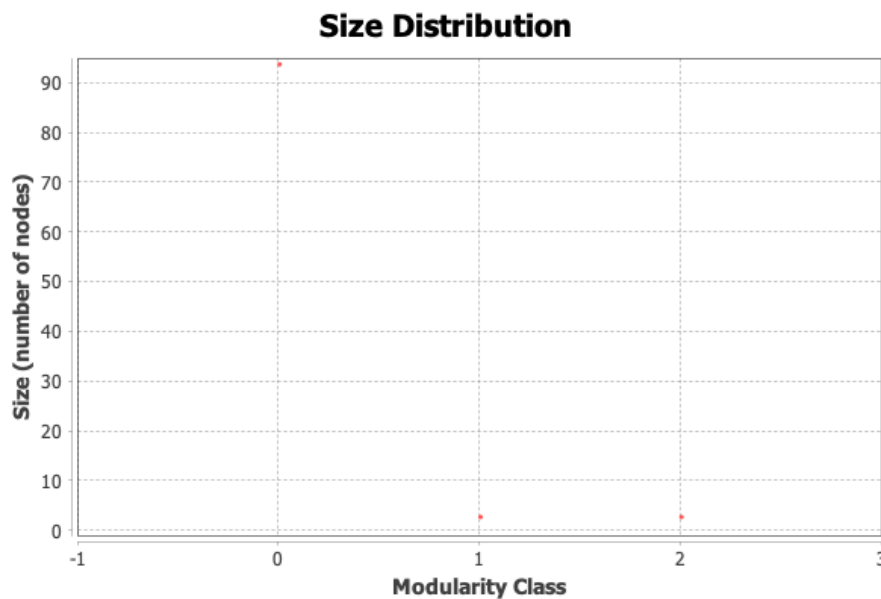


Figure 4.9: Community size distribution obtained from modularity analysis.

Figure 4.9 further supports this interpretation. One dominant community contains almost all nodes, while the remaining communities are extremely small.

These findings confirm that the graph does not naturally divide into several well-separated social groups. Instead, it behaves as a centralized interaction structure organized around dominant hubs.

4.14 Overall Results

The main results are summarized in Table 4.4. All analytical methods consistently point toward the same structural conclusion: the graph is sparse, fragmented, highly centralized, and structurally dependent on a small number of dominant actors.

Property	Result
Number of nodes	718
Number of edges	720
Density	0.0028
Average degree	2.01
Maximum degree	278
Number of connected components	60
Largest component share	79.9%
Maximum k-core number	3
Top Personalized PageRank node	Jeffrey Epstein
Top betweenness centrality node	Jeffrey Epstein
Modularity	0.082
Number of communities	3
Dominant structural pattern	Sparse hub-and-spoke / core-periphery topology

Table 4.4: Summary of the main network-analysis results.

The exploratory visualization first suggested strong centralization. The connected-component analysis then revealed one dominant giant component together with many peripheral fragments. The degree distribution showed the existence of extremely dominant hubs, while Personalized PageRank demonstrated that diffusion processes remain heavily concentrated around Jeffrey Epstein. Betweenness centrality further showed that shortest paths are also concentrated around the same central actor.

The k-core decomposition confirmed that the graph contains only a shallow structurally cohesive nucleus, while the robustness simulation demonstrated that the network rapidly collapses under targeted hub removal. Finally, modularity analysis showed that the graph does not exhibit strong decentralized community structure.

Taken together, these findings indicate that the communication graph is best interpreted

as a sparse, centralized, and structurally fragile interaction network rather than a balanced decentralized social network.

4.15 Limitations

Several limitations should be acknowledged.

First, the graph was constructed from automatically extracted entities and relationships, meaning that some noisy or ambiguous nodes remain present. Examples include email addresses or imperfectly resolved textual entities.

Second, the analysis depends strongly on the selected interaction types. Focusing on **sent email to**, **sent message to**, and **met with** produces a communication-centered graph. Alternative interaction types such as legal or financial relationships could potentially reveal different structural patterns.

Third, the analysis is structural rather than causal. Centrality measures do not imply guilt, responsibility, or factual real-world influence. They only describe structural positions within the extracted network.

Finally, Personalized PageRank is inherently seed-dependent. Since Jeffrey Epstein was selected as the seed node, the analysis specifically investigates the diffusion neighborhood surrounding this actor rather than a global ranking of all entities in the full dataset.

4.16 Conclusion

This analysis used graph-mining methods to investigate the structural organization of an Epstein-related communication network. Starting from exploratory graph visualization, the communication graph immediately appeared sparse and highly centralized. This observation motivated the use of graph diagnostics, Personalized PageRank, betweenness centrality, k-core decomposition, robustness simulations, and modularity analysis.

The results consistently show that the network is dominated by one highly central structural hub. Jeffrey Epstein obtained by far the highest Personalized PageRank score and also the highest betweenness centrality score. This indicates that he occupies both the dominant diffusion position and the dominant brokerage position inside the extracted communication structure.

The modularity analysis further supports this interpretation. With a modularity score of only 0.082, the graph does not exhibit strong decentralized community organization. In-

stead, it behaves as a hub-and-spoke communication network with many weakly connected peripheral nodes.

Overall, the analysis demonstrates that the extracted communication graph is best understood as a sparse, centralized, and structurally fragile interaction network rather than a decentralized community-based social structure.

Appendix A

Supplementary Material

A.1 Supplementary Material

Appendix B

Additional Visualizations

B.1 ConvNeXt Correct Heatmaps

This section presents additional correct prediction heat maps from ConvNeXt-Tiny, complementing the misclassification-focused examples discussed in Figure 2.6. Together, these examples provide a broader view of the regions that most strongly influence correct predictions across categories.



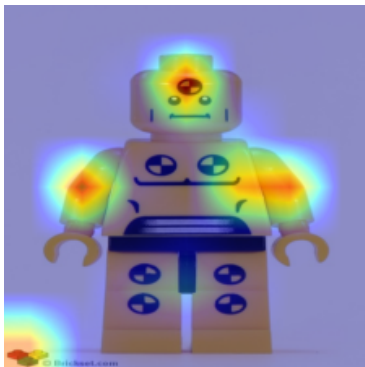
Belville



BrickLink Designer Program



Castle



Collectible Minifigures



DREAMZzz



DUPLO

Figure B.1: Additional correct prediction heat maps from ConvNeXt-Tiny, part I.

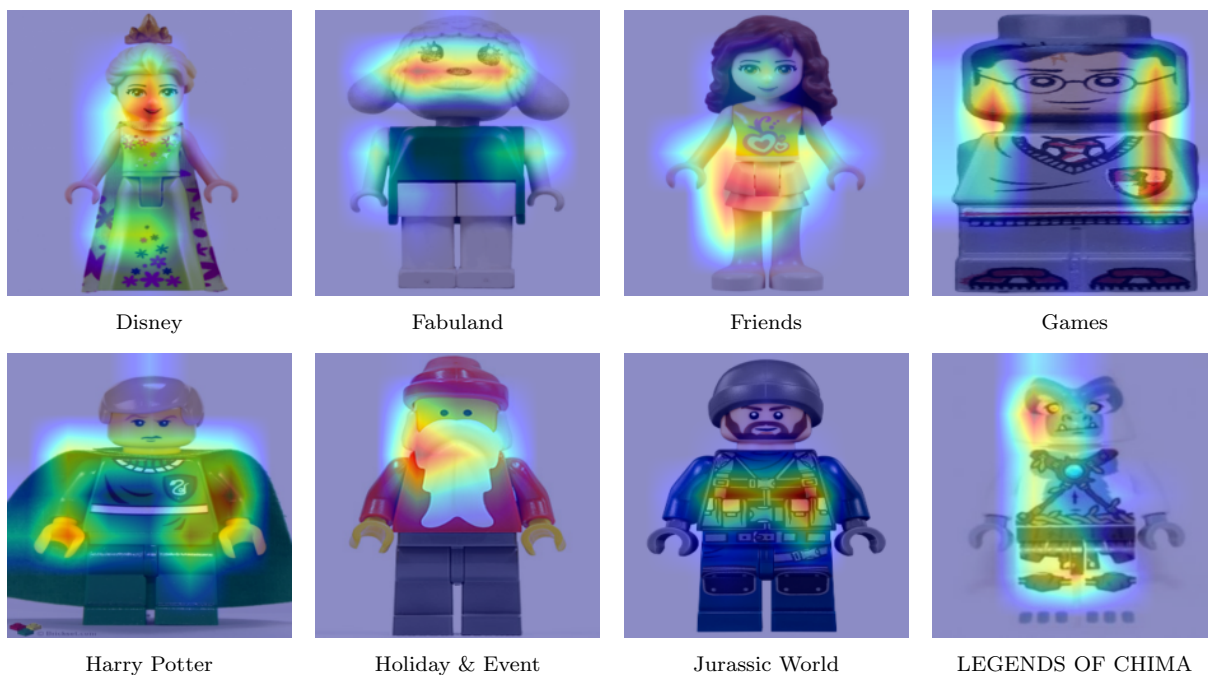


Figure B.2: Additional correct prediction heat maps from ConvNeXt-Tiny, part II.

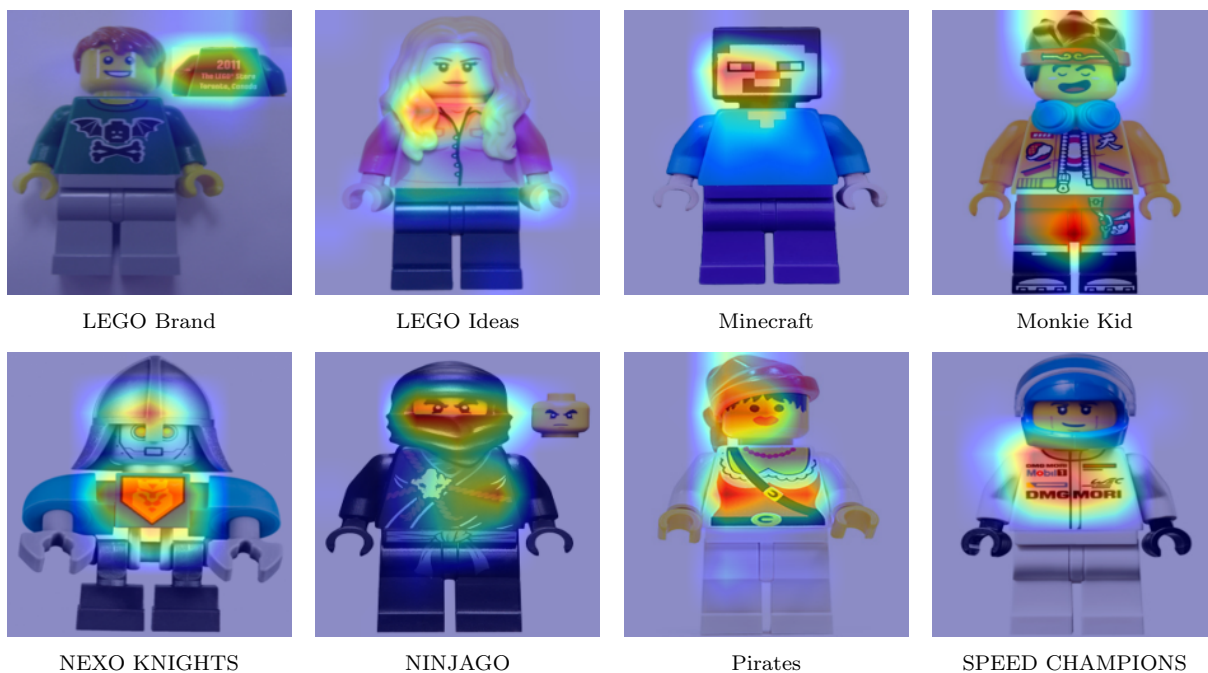


Figure B.3: Additional correct prediction heat maps from ConvNeXt-Tiny, part III.

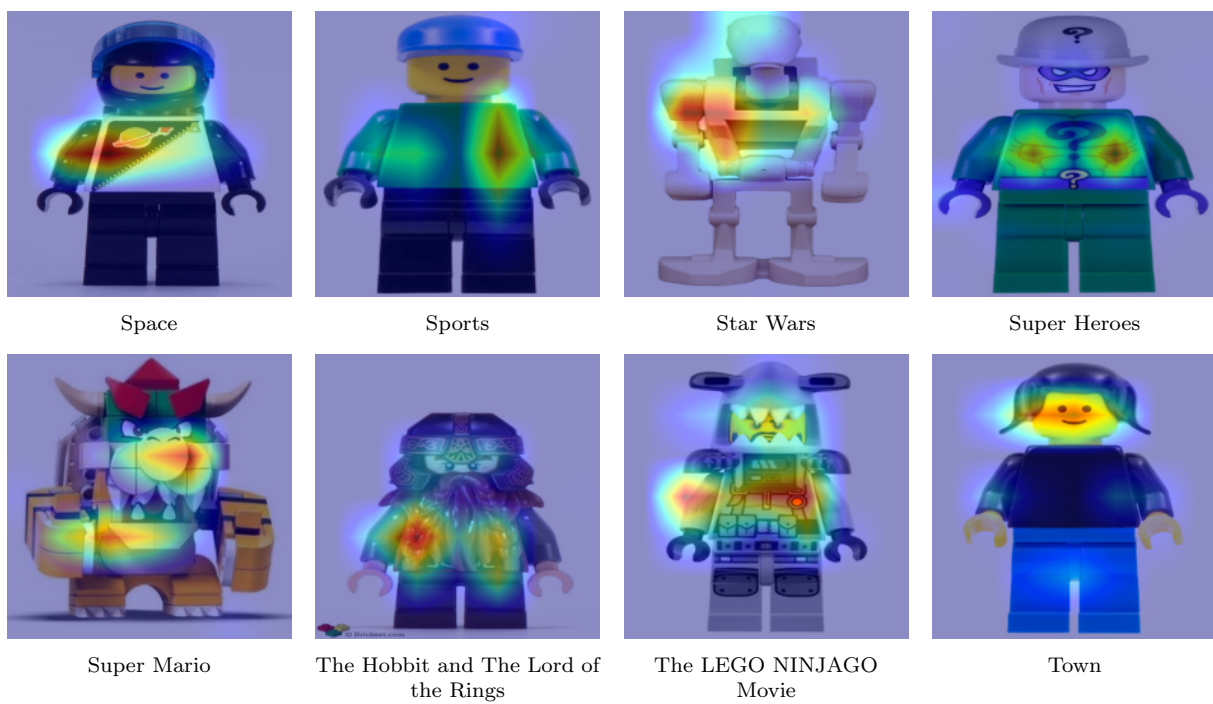


Figure B.4: Additional correct prediction heat maps from ConvNeXt-Tiny, part IV.

B.2 Confusion Matrices

This section collects the test-set confusion matrices for all base models, fusion models, and ensemble variants considered in the study.

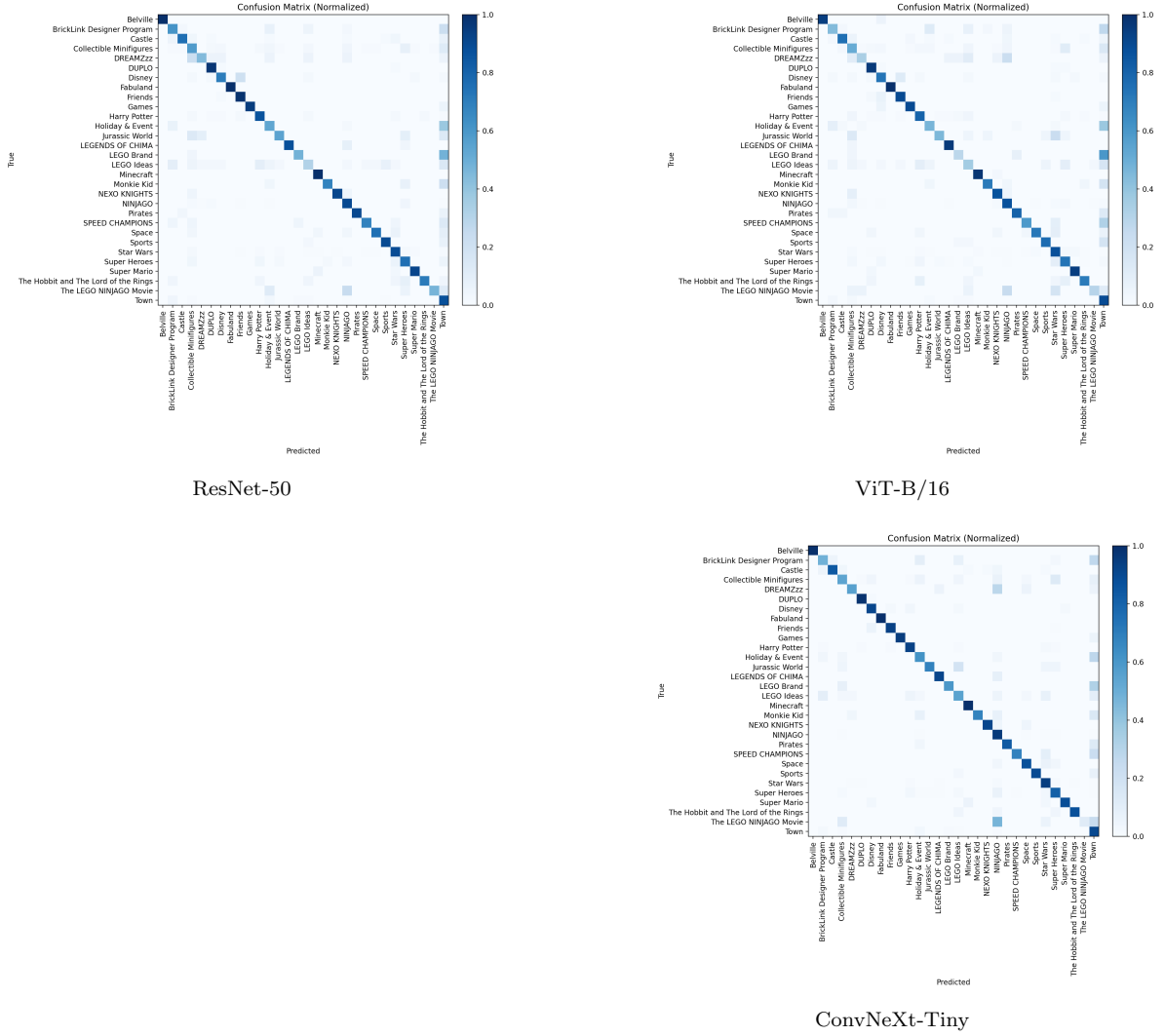
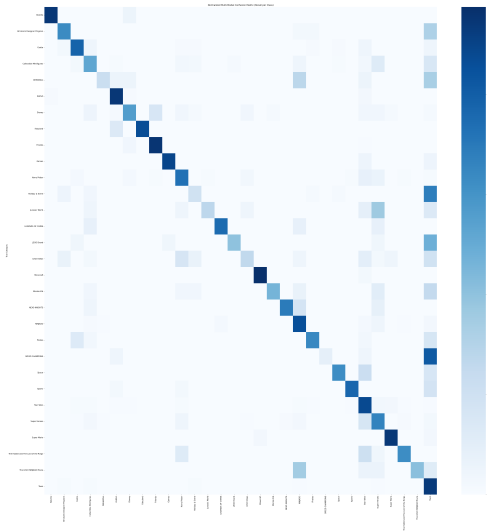
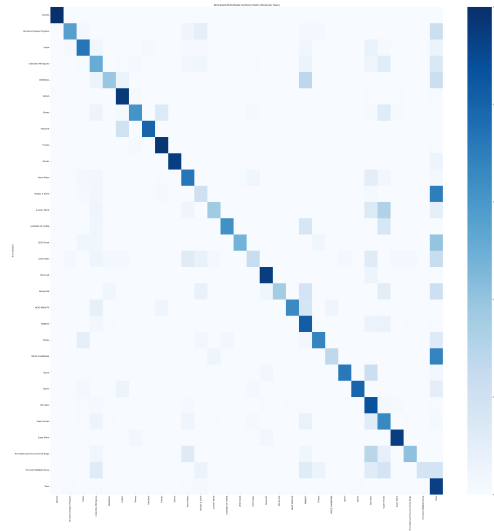


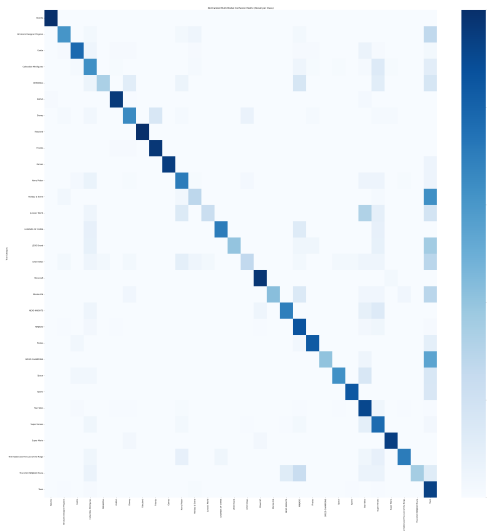
Figure B.5: Test-set confusion matrices for the fine-tuned single-modality deep learning models.



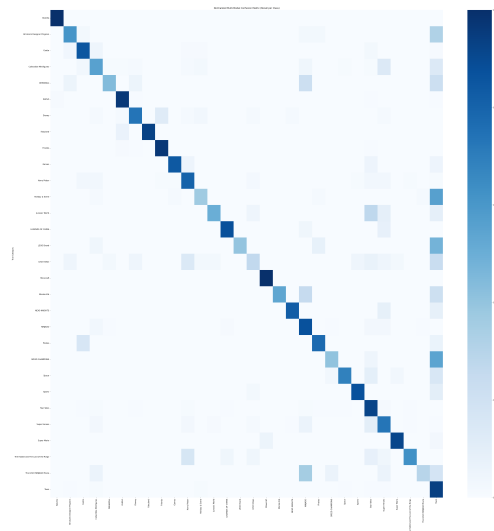
XGBoost + ResNet-50 features



XGBoost + ViT-B/16 features

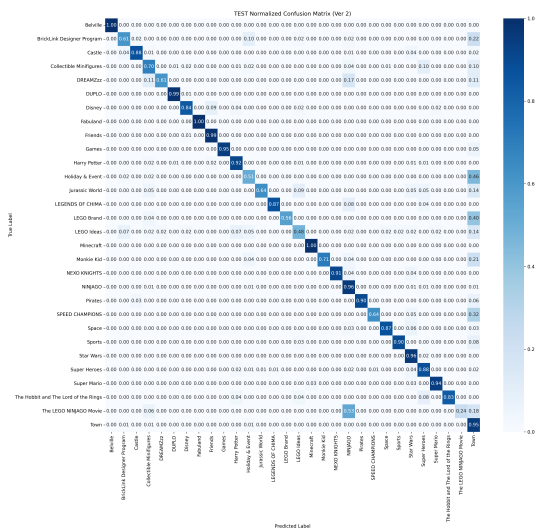


XGBoost + DINOv2 features

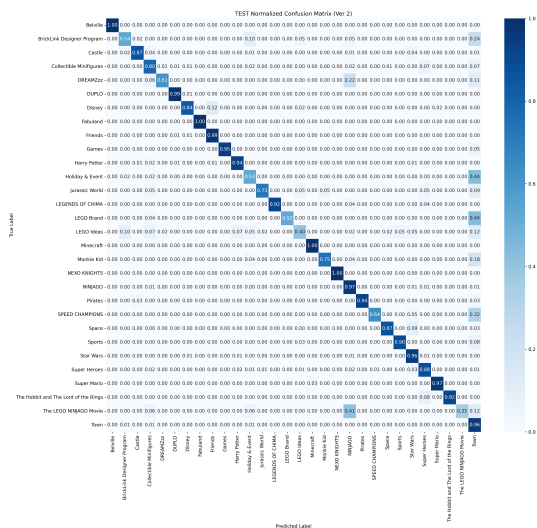


XGBoost + CLIP features

Figure B.6: Test-set confusion matrices for the metadata-augmented XGBoost fusion models.



Ensemble base (no TTA)



Ensemble + TTA ($N = 8$)

Figure B.7: Test-set confusion matrices for the ensemble models.